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DNA Methylation Dynamics Along the Normal–Adjacent Axis in Breast Cancer

A Quantitative Statistical and Machine Learning Approach

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3.1 Methodological Objectives

This chapter establishes the methodological foundation for all subsequent analyses by addressing the construction, organization, and preliminary characterization of the DNA methylation datasets used in this thesis. Following the biological background and problem formulation introduced in Chapters 1 and 2, the present chapter acts as a bridge between domain-specific knowledge and quantitative modeling, ensuring that downstream analyses are grounded on a technically sound and well-understood data representation.

Genome-wide DNA methylation data are characterized by extremely high dimensionality, heterogeneous signal-to-noise ratios, and the coexistence of subtle biological effects with multiple sources of technical variability. In this context, premature application of statistical models or feature-selection procedures may lead to biased conclusions driven by artefacts rather than genuine biological structure. A careful examination of data quality, distributional properties, and internal consistency is therefore a necessary prerequisite for any reliable inference [1].

The primary objective of this chapter is to construct a coherent and reproducible representation of the available datasets and to assess their structural and statistical properties prior to any form of preprocessing or modeling. Specifically, this chapter aims to: (i) describe the organization of methylation matrices and associated phenotype information; (ii) characterize the global and local variability of methylation levels within each dataset; (iii) evaluate the degree of separation between Normal, Adjacent, and Tumor tissues using exploratory tools; and (iv) compare datasets at an inter-cohort level to assess consistency, heterogeneity, and potential limitations in signal transferability.

The scope of this chapter is deliberately restricted to exploratory and diagnostic analyses. No feature selection, predictive modeling, or formal statistical inference is performed at this stage. All analyses are intended to be descriptive and comparative, serving to highlight structural properties of the data and to identify potential technical or biological issues that must be addressed before proceeding further.

Methodologically, the chapter is organized around a unified exploratory framework applied consistently across all datasets. Intra-dataset analyses are used to investigate internal structure, variability patterns, and group-level behavior, while inter-dataset comparisons focus on assessing cross-cohort consistency and platform-related differences. Throughout the chapter, visualizations and summary metrics are employed as diagnostic tools rather than as decision-making instruments.

The results of this exploratory phase reveal both shared characteristics and dataset-specific behaviors, as well as clear limitations of working directly with raw or minimally processed methylation data. These observations motivate the need for a structured and rigorous data pre-processing pipeline, which is developed and justified in the following chapter. The present chapter therefore provides the conceptual and technical basis for the preprocessing choices adopted in Chapter 4 and for all subsequent modeling steps.

3.2 Dataset Construction, Metadata Integration, and Data Organization

This section provides the methodological framework for the selection, construction of the dataset, and the organisation process adopted in this study. It contextualises the data sources, representation choices, metadata management, and computational design principles that underpin the detailed implementation described in the following sections.

3.2.1 Data Sources and Study Design

All datasets analyzed in this thesis were obtained from the NCBI Gene Expression Omnibus (GEO) [2], which was selected as a unified data source to minimize heterogeneity in data formats, metadata conventions, and access mechanisms. GEO provides curated series-level accessions, stable sample identifiers, and standardized links to processed methylation matrices and sample-level annotations, making it a natural reference repository for large-scale DNA methylation studies.

The study design adopts a multi-cohort comparative framework focused on genome-wide DNA methylation in breast tissue. Three independent GEO series were selected for analysis: **GSE69914** [3], based on the Illumina HumanMethylation450 platform, and two more recent EPIC-based cohorts, **GSE225845** [4] and

GSE287331 [5]. Together, these datasets span normal/healthy, adjacent-normal, and tumor tissue states across two generations of Illumina methylation arrays (450K and EPIC), enabling both within-dataset characterization and cross-dataset comparisons.

Dataset selection was driven by strict inclusion criteria aimed at ensuring technical compatibility, biological relevance, and analytical feasibility. In particular, only studies providing processed genome-wide methylation matrices in terms of β-values were considered, allowing a uniform data representation and avoiding the need for low-level signal reprocessing from raw `.idat` intensity files. Raw `.idat` files store probe-level fluorescence intensities measured in the methylated ($y^{(M)}$) and unmethylated ($y^{(U)}$) channels, from which β-values are derived as a normalized ratio [6]:

$$\beta := \frac{\max(y^{(M)},0)}{\max(y^{(M)},0) + \max(y^{(U)},0) + \alpha}, \qquad \beta \in [0,1]. \tag{3.1}$$

where α is a small offset (typically 100) used to stabilize the denominator. Additional requirements included the use of Illumina HumanMethylation450 (450K) or MethylationEPIC (EPIC) platforms, the availability of multiple tissue classes within each cohort, and sufficiently large sample sizes to support robust statistical analysis.

3.2.2 Methylation Data Representation and Storage

In several datasets, the original authors applied established normalization procedures, including methods specifically designed to correct probe-type design bias such as BMIQ [7]. Reprocessing raw IDAT files would therefore not add information relevant to the objectives of this thesis, while substantially increasing technical complexity and reducing reproducibility. For these reasons, all analyses were conducted on the processed β-value matrices as distributed via GEO.

To support scalable analysis and efficient input/output operations, all working methylation matrices were converted to a standardized `sample` × `CpG` layout and stored in columnar Parquet format using single-precision floating-point encoding.

This choice is motivated by both numerical and computational considerations. Since β-values are strictly bounded in the interval [0,1] [6], and biologically meaningful methylation differences typically occur at magnitudes on the order of 10^{-2}–10^{-3}, single-precision floating point (`float32`, machine $\varepsilon \approx 10^{-7}$) provides more than sufficient numerical accuracy. At the same time, it substantially reduces memory usage and I/O overhead compared to double precision.

This representation is consistent with recent large-scale genomics frameworks that process molecular features, including DNA methylation data, entirely in `float32` precision for efficiency and scalability [8].

3.2.3 Phenotype Tables and Label Harmonization

For each dataset, a dedicated phenotype table was constructed by extracting and parsing GEO sample-level metadata. These tables encode tissue labels, sample identifiers, and relevant clinical or technical covariates when available. Given the heterogeneity of original annotations across studies, tissue classes were mapped to harmonized numeric encodings to ensure consistency across cohorts while preserving dataset-specific attributes in separate fields.

A strict one-to-one correspondence between methylation matrices and phenotype tables was enforced. Samples lacking either methylation measurements or corresponding metadata were excluded to prevent inconsistencies and downstream misalignment. This design guarantees that all analyses operate on synchronized molecular and phenotypic representations and enables reliable stratification by tissue class in both intra- and inter-dataset comparisons.

3.2.4 Reproducibility and Computational Workflow

The overall data organization and computational design follow a multi-dataset analytical paradigm commonly adopted in recent large-scale DNA methylation studies, in which cohort-specific analyses are complemented by cross-dataset comparisons to explicitly evaluate robustness and generalizability, while simultaneously revealing cohort-dependent effects and limitations in signal transferability that motivate subsequent methodological refinement [9].

All dataset construction and organization steps were implemented using deterministic, script-based pipelines with explicit schema definitions and typed data representations. Particular care was taken to avoid unnecessary in-memory operations on high-dimensional methylation matrices, favoring streaming ingestion, chunked processing, and out-of-core transformations when required by dataset size.

The resulting data organization yields a set of standardized, self-contained datasets, each consisting of a genome-wide methylation matrix and an aligned phenotype table stored in efficient, portable formats. This design ensures full reproducibility of the data preparation process and provides a robust foundation for the preprocessing, exploratory analysis, and modeling steps developed in subsequent chapters.

3.3 Intra-Dataset Exploration and Visualization

This section presents a structured intra-dataset exploratory analysis aimed at characterising the internal statistical and biological organisation of each methylation cohort prior to any preprocessing or modelling step. Each dataset is analysed independently in order to assess data integrity, quantify variability patterns, and

Table 3.1: Overview of the GSE69914 dataset after initial cohort curation.

CpGs	Samples	Normal	Adjacent	Tumor	Normal (BRCA1)	Tumor (BRCA1)
485,512	407	50	42	305	3	7

evaluate whether Normal, Adjacent, and Tumor tissues exhibit distinguishable epigenetic behaviour.

In the context of breast cancer, epigenetic deregulation is known to manifest through both global and local phenomena, including genome-wide hypomethylation, focal hypermethylation of tumour-suppressor regions, and the presence of pre-neoplastic "field defects" in histologically normal tissues proximal to tumours [10, 11, 12]. These mechanisms motivate an exploratory investigation focused not only on overt tumour-associated changes, but also on more subtle alterations in Normal and Adjacent samples, which are central to the biological hypothesis of this thesis.

This section therefore applies a unified exploratory framework to each dataset (GSE69914 [3.3.1], GSE225845 [3.3.2], GSE287331 [3.3.3]), systematically examining data quality, global methylation distributions, sample-level summaries, CpG-wise variability and instability, correlation structure, dimensionality reduction, and genomic context coverage. The use of a consistent analytical protocol enables direct comparison of intra-dataset behaviour while preserving cohort-specific characteristics.

3.3.1 Dataset GSE69914

Dataset overview GSE69914 is a breast tissue DNA methylation cohort profiled using the Illumina Infinium HumanMethylation450 BeadChip, providing genome-wide coverage of approximately 4.8×10^5 CpG loci. The dataset comprises samples spanning the full Normal–Adjacent–Tumor spectrum that motivates the biological framework of this thesis.

An overview of the sample composition and CpG coverage is reported in Table 3.1, including the distribution across tissue classes and hereditary-risk subgroups.

A limited subset of samples is associated with germline *BRCA1* mutations. *BRCA1*, in particular, encodes a key DNA repair protein, and pathogenic variants are known to substantially increase lifetime breast and ovarian cancer risk [13]. Multiple studies have shown that BRCA-associated breast tissues exhibit distinct baseline epigenetic profiles, reflecting inherited genomic instability rather than sporadic tumorigenesis [14, 15].

Since the primary objective of this thesis is to characterise early, field defect DNA methylation alterations along the Normal–Adjacent axis in sporadic breast cancer, samples carrying *BRCA1* mutations were excluded from the core exploratory and

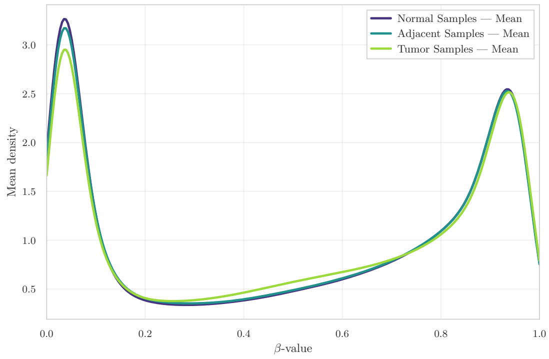


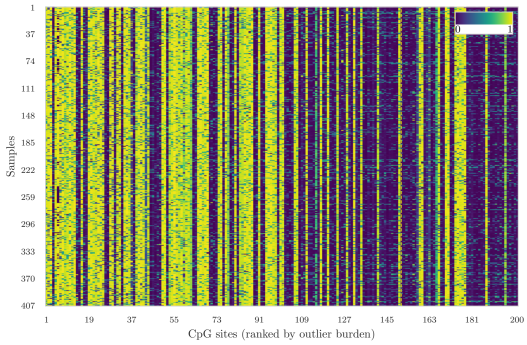
Figure 3.1: Group-wise mean β-value density in GSE69914.

preprocessing pipeline. Their inclusion would introduce a confounding hereditary epigenetic signal, potentially obscuring subtle methylation shifts in non-mutated normal and adjacent tissues.

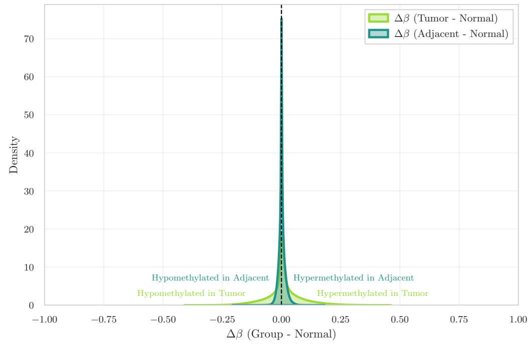
Global methylation distributions To characterise the overall methylation landscape of the dataset, the distribution of β-values (fractional methylation in [0,1]) was first examined. The group-wise mean density curve (Figure 3.1) displays the characteristic *bimodal* profile of Illumina 450k arrays, with peaks near unmethylated ($\beta \approx 0$) and fully methylated ($\beta \approx 1$) CpG sites. This pattern reflects the underlying biology of CpG regulation, where many loci tend to be either transcriptionally active (hypomethylated) or repressed (hypermethylated) [11, 16].

When comparing tissue groups, a clear gradient emerges. Tumor samples show a slightly flatter high-β peak and a broader low-β tail, consistent with the well-described phenomenon of global hypomethylation and increased heterogeneity in cancer [10]. Adjacent samples lie between Normal and Tumor, suggesting early epigenetic drift and subtle field defects occurring in histologically non-neoplastic tissue [12].

CpG-level instability and recurrent outliers To characterise locus-specific instability, the 200 CpG sites with the highest outlier burden across samples were examined. The heatmap of the top outlier loci (Figure 3.2a) indicates that epigenetic disruption is *not uniform*: specific CpG sites and specific samples display extreme deviations, rather than a diffuse genome-wide shift. Moreover, both directions of alteration are present — focal hypermethylation and focal hypomethylation. In cancer biology, hypermethylation can silence tumor-suppressor regions, whereas hypomethylation can derepress oncogenic pathways and weaken genomic stability, reflecting the classic "too much and too little methylation" behaviour of tumor genomes [10].



(a) Top outlier CpG loci.



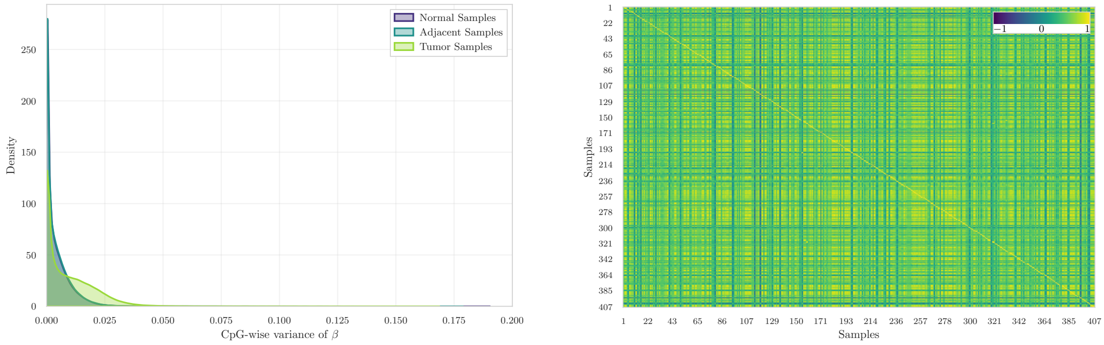
(b) $\Delta\beta$ distributions (Tumor/Adjacent − Normal).

Figure 3.2: Heatmap of the top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right), illustrating CpG-level instability in GSE69914.

Complementary $\Delta\beta$ distributions comparing Tumor and Adjacent tissues against Normal (Figure 3.2b) show a clear shift toward hypomethylation in Tumor samples and a subtler but detectable drift in Adjacent tissues. This provides evidence that epigenetic alterations emerge early in histologically normal tissue located near the tumor, supporting the field-cancerization model [12].

Variance and correlation structure To quantify within-group epigenetic variability, the distribution of CpG-wise variance across samples was examined. The density curves (Figure 3.3a) show that Tumor samples have markedly higher variance, reflecting increased epigenetic instability. In contrast, Normal and Adjacent samples display almost identical variance distributions that are tightly concentrated near zero. This indicates that, at the level of CpG-wise variability, Adjacent tissues do not yet exhibit detectable divergence from Normal samples.

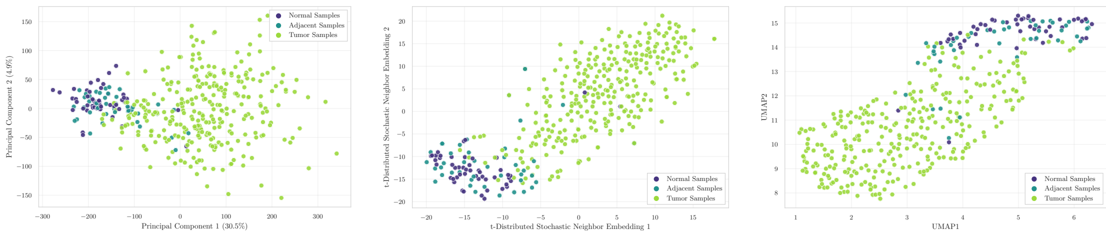
A complementary perspective is provided by the sample correlation heatmap (Figure 3.3b), which shows the expected high level of pairwise similarity across all samples. This behaviour is typical of genome-wide DNA methylation data, where a large fraction of CpG sites is stable across individuals, resulting in consistently strong correlations [7]. Importantly, the heatmap also reveals that all samples are highly mutually correlated, with no sharp blocks or boundaries separating the three tissue labels. This confirms that global methylation structure is largely conserved across Normal, Adjacent and Tumor tissues, and that group differences are too subtle to be detected through sample–sample correlation alone.



(a) CpG-wise variance distributions.

(b) Sample correlation heatmap.

Figure 3.3: Variance and correlation structure across Normal, Adjacent, and Tumor samples in GSE69914.



(a) PCA trustworthiness = 0.869.

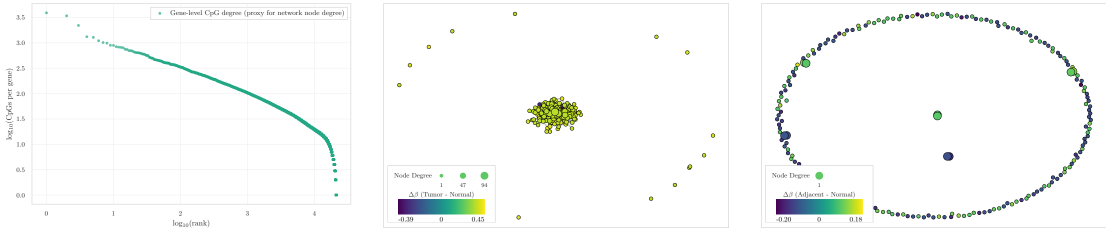
(b) t-SNE trustworthiness = 0.888.

(c) UMAP trustworthiness = 0.874.

Figure 3.4: Low-dimensional embeddings samples in GSE69914.

Low-dimensional embeddings Dimensionality reduction was applied using Principal Component Analysis (PCA), t-distributed Stochastic Neighbor Embedding (t-SNE), and Uniform Manifold Approximation and Projection (UMAP) to visualise global similarities among samples (Figure 3.4). Across all three methods, Normal and Adjacent samples show substantial overlap, with no clear separation between these two groups. Tumor samples appear more dispersed, but only mildly shifted relative to the Normal/Adjacent cluster, and no sharp cluster boundaries emerge in any embedding. The trustworthiness scores are similar across embeddings (0.869–0.888), indicating that local neighbourhood relationships are largely preserved in the low-dimensional projections.

These projections indicate that global methylation patterns alone do not provide strong discriminative structure among the three tissue states. This suggests that group differences are subtle at the whole-methylome level and are more effectively captured through locus-specific analyses rather than through global unsupervised embeddings.



(a) Gene-level CpG degree distribution.

(b) DNB-like network: Tumor vs Normal.

(c) DNB-like network: Adjacent vs Normal.

Figure 3.5: Dynamic-network proxy visualizations in GSE69914.

Genome annotation coverage CpG probes were annotated using the official *Infinium MethylationEPIC v1.0 B5* [17] manifest file, the standard Illumina resource containing genomic coordinates, probe type, CpG island context (Island, Shore, Shelf, Open Sea), and gene-level annotations. Using this manifest, the dataset exhibits the expected non-uniform genomic distribution of CpG contexts: the largest fraction of probes maps to open-sea or non-island regions, followed by CpG islands, then north and south shores, and finally north and south shelves. This ordering mirrors the characteristic architecture of the HM450 arrays reported in the literature [18, 19]. The coverage observed is therefore consistent with the known genomic composition of the Illumina methylation arrays, supporting correct manifest alignment.

Dynamic-network proxies Although this analysis does not follow the full mathematical formalism of Dynamic Network Biomarkers (DNBs), the degree-based and network-level (Figure 3.5) provide a qualitative view of how methylation perturbations may organize at the network level. DNB theory, originally formulated for gene-expression and regulatory networks, predicts that near a critical transition a subset of components displays increased internal fluctuations and coordinated behaviour [20, 21, 22]. Here, this conceptual framework is adapted to CpG-level methylation patterns by examining the structure of CpG–gene degree distributions and the organization of $\Delta\beta$-driven network layouts. This approach enables the exploration of whether tumour-associated instability manifests as locally coordinated methylation shifts at the network level, even without implementing the complete DNB pipeline.

The degree–rank plot (Figure 3.5a) displays a heavy-tailed distribution of CpG counts per gene, consistent with the heterogeneous hub–periphery structure typically observed in biological systems. In the Tumor vs Normal comparison (Figure 3.5b), a compact nucleus of nodes with moderately high degree is observed, accompanied by larger $\Delta\beta$ deviations. This pattern is consistent with a more perturbed and locally coordinated subset of CpG/gene units in tumor tissue, qualitatively compatible with increased network instability. In contrast, the Adjacent vs Normal network (Figure 3.5c) does not exhibit a comparable hub-like concentration and shows smaller, more diffuse deviations, suggesting that the adjacent tissue reflects only early or weakly organized perturbations.

Although these observations do not satisfy the full mathematical criteria for identifying a DNB module, they are conceptually consistent with the interpretation that tumor tissue occupies a more unstable network regime, whereas adjacent tissue remains closer to a stable (pre-transition) configuration.

Overall interpretation The exploratory analysis of GSE69914 reveals a consistent but overall subtle separation between the three tissue states. Normal and Adjacent samples exhibit highly similar global methylation profiles, variance distributions, correlation structure, and low-dimensional embeddings, indicating that large-scale methylome organisation remains largely conserved in Adjacent tissue. Across these global summaries, the two groups are so intermixed that no clear or marked distinction emerges between Normal and Adjacent samples when using whole-methylome descriptors. Tumor samples display the expected increase in heterogeneity and a shift toward hypomethylation; however, these differences remain modest at the whole-methylome level and do not produce clear separation in unsupervised embeddings.

Locus-specific analyses, however, highlight focused patterns of disruption. Outlier CpGs exhibit both hyper- and hypomethylation events, and $\Delta\beta$ distributions reveal a distinct hypomethylation bias in Tumor samples together with a milder but detectable shift in Adjacent tissue. Dynamic-network proxies further indicate a more compact and perturbed core in Tumor samples, whereas Adjacent tissue appears more diffuse and weakly structured.

Overall, GSE69914 constitutes a technically clean dataset with well-structured methylation patterns and biologically interpretable deviations. The results indicate that tumour-related alterations are primarily localized at specific CpG loci rather than reflected in global methylome reorganization. Within this framework, the Adjacent tissue exhibits subtle yet measurable deviations from the Normal baseline, consistent with previously described field-effect patterns in breast cancer epigenomics.

3.3.2 Dataset GSE225845

Dataset overview GSE225845 is a breast tissue DNA methylation cohort profiled using the Illumina Infinium MethylationEPIC BeadChip (850K), providing genome-wide coverage of more than 8.5×10^5 CpG loci. The dataset is part of the

Table 3.2: Overview of the GSE225845 dataset after initial cohort curation.

CpGs	Samples	Normal	Adjacent	Tumor
750,426	477	113	140	224

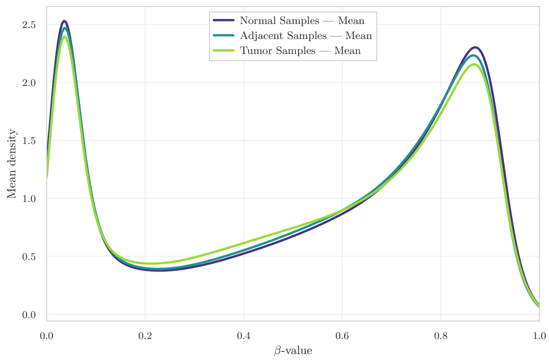


Figure 3.6: Group-wise mean β-value density in GSE225845.

NCI-Maryland Breast Cancer Cohort and includes samples spanning the Normal–Adjacent–Tumor spectrum that motivates the biological framework of this thesis.

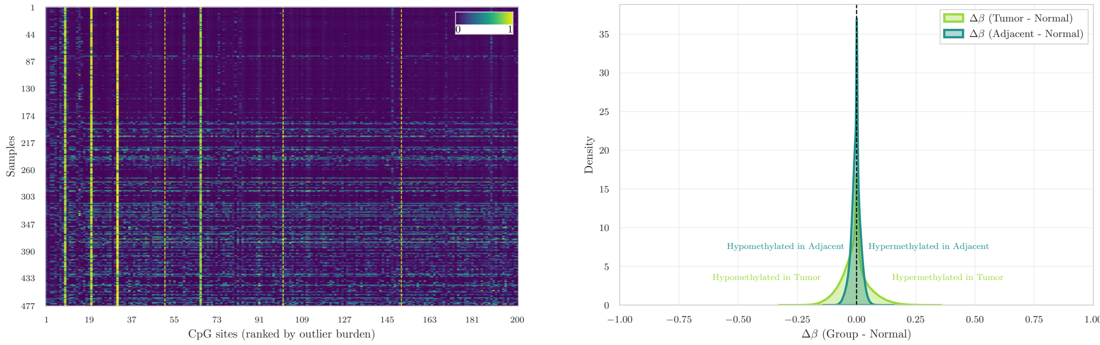
An overview of the sample composition and CpG coverage after cohort harmonization is reported in Table 3.2, including the distribution across tissue classes.

The dataset further provides detailed demographic and clinical metadata (e.g., age at surgery, race, sex, and tissue descriptors), enabling controlled downstream analyses when required.

Global methylation distributions The group-wise mean β-value densities (Figure 3.6) display the expected bimodal configuration of EPIC arrays. As observed in GSE69914, the three tissue groups show a high degree of overlap across the entire β range.

Normal and Adjacent samples exhibit nearly indistinguishable density profiles. Tumor samples present only minimal deviations, characterised by a slight increase in density at low β values and a marginal attenuation of the high-β peak. At the whole-methylome level, no marked global redistribution of methylation levels is evident.

CpG-level instability and recurrent outliers The heatmap of the top outlier CpG loci (Figure 3.7a) indicates that instability within this subset is predominantly directional. Recurrent outlier events are largely associated with very low β values, consistent with focal hypomethylation affecting a restricted set of CpG sites. The



(a) Top outlier CpG loci.

(b) $\Delta\beta$ distributions (Tumor/Adjacent − Normal).

Figure 3.7: Heatmap of the top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right), illustrating CpG-level instability in GSE225845.

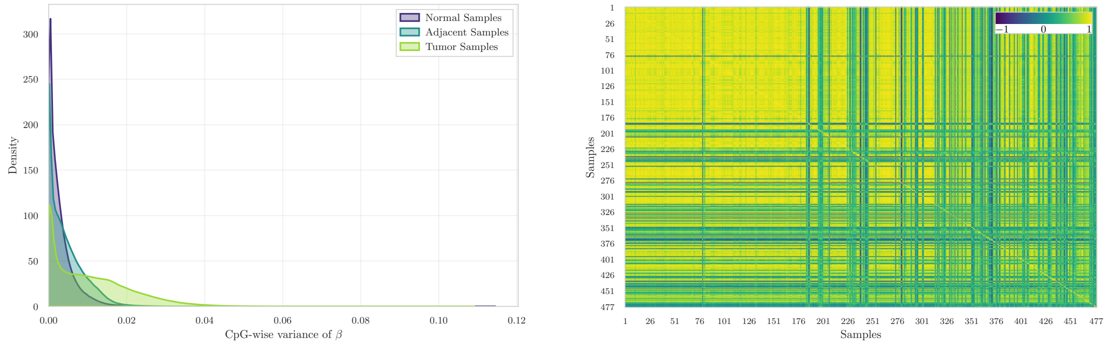
perturbation pattern therefore appears concentrated rather than diffuse, suggesting localized instability rather than a genome-wide redistribution.

The corresponding $\Delta\beta$ distributions comparing Tumor and Adjacent tissues against Normal (Figure 3.7b) are sharply centered around zero, indicating that most loci remain stable. However, both comparisons exhibit a heavier negative tail relative to the positive side. Tumor samples display the most extended negative tail, consistent with stronger hypomethylation shifts, whereas Adjacent tissue shows a milder but still detectable asymmetry toward $\Delta\beta < 0$.

Variance and correlation structure The CpG-wise variance distributions (Figure 3.8a) indicate clear differences in variability across tissue groups. Normal samples exhibit highly concentrated variance values near zero, whereas Tumor samples display a substantially longer right tail, reflecting an increased proportion of CpG loci with elevated variability. Adjacent samples show a broader distribution than Normal tissues, with partial overlap with both Normal and Tumor profiles, but without forming a strictly intermediate pattern.

The sample correlation heatmap (Figure 3.8b) shows uniformly positive correlations across the dataset. Most pairwise values lie within a narrow high-correlation range, and no distinct block structures separating tissue classes are observed. This pattern indicates preservation of global methylation structure, with group differences emerging primarily from locus-specific variability rather than from large-scale shifts in overall methylome organization.

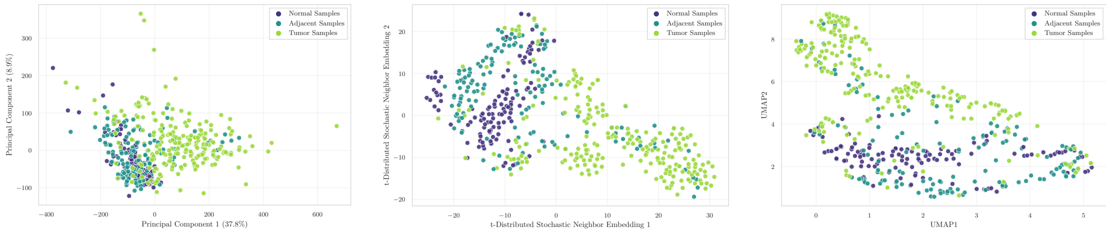
Low-dimensional embeddings Dimensionality reduction was applied using PCA, t-SNE, and UMAP to visualise global similarities among samples (Figure 3.9).



(a) CpG-wise variance distributions.

(b) Sample correlation heatmap.

Figure 3.8: Variance and correlation structure across Normal, Adjacent, and Tumor samples in GSE225845.



(a) PCA trustworthiness = 0.881.

(b) t-SNE trustworthiness = 0.939.

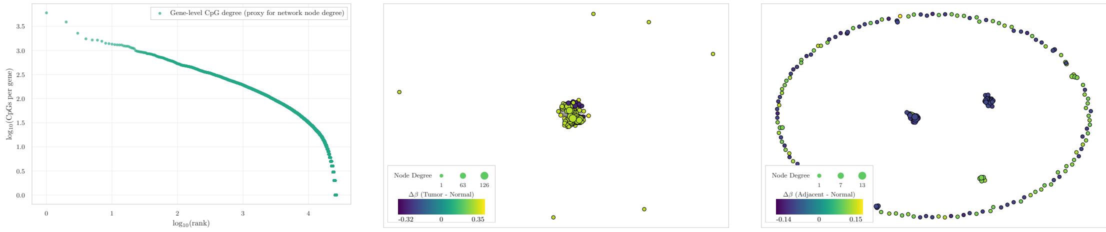
(c) UMAP trustworthiness = 0.894.

Figure 3.9: Low-dimensional embeddings samples in GSE225845.

In PCA space, the three tissue groups largely overlap. Tumor samples show a slightly broader dispersion, particularly along the second principal component, but no clear linear separation is observed.

Nonlinear embeddings reveal additional structure. In t-SNE space, Tumor samples occupy a more extended region and appear more dispersed, whereas Normal and Adjacent samples remain partially intermixed. Several Tumor samples are positioned at the periphery of the embedding, consistent with increased heterogeneity.

The UMAP projection exhibits a similar configuration. Tumor samples span a wider area and extend toward higher values of the second UMAP axis, while Normal and Adjacent samples form relatively compact yet overlapping regions. Trustworthiness scores range from 0.881 to 0.939 across embeddings, indicating good preservation of local neighbourhood structure, particularly in the nonlinear projections. Despite differences in spread, none of the methods yields sharply separated or isolated clusters.



(a) Gene-level CpG degree distribution. **(b)** DNB-like network: Tumor vs Normal. **(c)** DNB-like network: Adjacent vs Normal.

Figure 3.10: Dynamic-network proxy visualizations in GSE225845.

Genome annotation coverage Consistent with the architecture of the EPIC platform, the genomic distribution of annotated CpG contexts is markedly unbalanced. The largest fraction of probes maps to open-sea or non-island regions, which constitute the dominant category. CpG islands represent the second most abundant class, followed by north and south shores in comparable proportions, whereas north and south shelves account for the smallest fractions.

This annotation profile aligns with the expected genomic composition of the EPIC array and supports correct manifest integration.

Dynamic-network proxies The degree–rank distribution (Figure 3.10a) exhibits a heavy-tailed profile, with a limited number of genes associated with many CpGs and a long tail of low-degree nodes. Such right-skewed degree structures are characteristic of heterogeneous biological networks [23].

In the Tumor vs Normal layout (Figure 3.10b), nodes aggregate into a dense and highly compact central cluster, with only a small number of isolated peripheral points. This geometry indicates that the largest Tumor–Normal $\Delta\beta$ deviations are concentrated within a cohesive subset of CpG–gene units rather than broadly distributed across the network.

In contrast, the Adjacent vs Normal layout (Figure 3.10c) displays a markedly different configuration. While a few small central aggregates are visible, most nodes are arranged along a wide annular structure, forming a pronounced ring-like embedding. This organisation reflects weaker and more spatially dispersed deviations, without the central concentration observed in the Tumor comparison.

Altogether, the proxy visualisations indicate that Tumor–Normal contrasts generate a compact and locally coherent network signal, whereas Adjacent–Normal differences remain diffuse and geometrically fragmented.

Overall interpretation The exploratory analysis of GSE225845 indicates that, at the whole-methylome level, the three tissue states share highly similar global

Table 3.3: Overview of the GSE287331 dataset after initial cohort curation.

CpGs	Samples	Normal	Adjacent	Tumor	OQ	CUB
866,552	446	182	60	69	67	68

methylation profiles. Group-wise mean β-value densities are nearly superimposed, and both correlation structure and linear embeddings show substantial overlap. These observations indicate preservation of large-scale methylome organisation across Normal, Adjacent, and Tumor tissues, without evidence of pronounced genome-wide redistribution of methylation levels.

Locus-focused analyses reveal more structured differences. Among the 200 CpGs with the highest outlier burden, instability is predominantly directional and characterised by focal hypomethylation. Variance distributions further show increased dispersion in Tumor samples, while Adjacent tissues exhibit broader variability than Normal without forming a strictly intermediate pattern.

Nonlinear embeddings (t-SNE and UMAP) emphasise differences in spatial dispersion rather than discrete cluster separation. Normal and Adjacent samples occupy relatively compact regions, whereas Tumor samples extend over a wider area, reflecting increased heterogeneity. Dynamic-network proxies show that Tumor–Normal contrasts generate a dense central cluster of perturbed CpG–gene units, while Adjacent–Normal differences are distributed along a broader, ring-like structure without strong central concentration.

Despite the larger sample size of this cohort, which provides stable estimates of global methylation structure, the separation between Normal and Adjacent tissues remains modest. Differences between these two groups are therefore primarily focal and do not manifest as large-scale structural shifts.

Overall, GSE225845 represents a globally homogeneous dataset in which tissue-state differences become apparent only when examined through locus-specific perturbations, variability patterns, and small-scale network organisation.

3.3.3 Dataset GSE287331

Dataset overview GSE287331 is a breast tissue DNA methylation cohort profiled using the Illumina Infinium MethylationEPIC v1.0 BeadChip. The dataset is organised along a tumor proximity axis (TPxA) spanning multiple anatomical and pathological contexts.

Five tissue categories are represented: healthy donated breast (HDB), contralateral unaffected breast (CUB), ipsilateral opposite quadrant (OQ), adjacent normal tissue (AN), and tumor (TU). An overview of the full cohort composition is reported in Table 3.3.

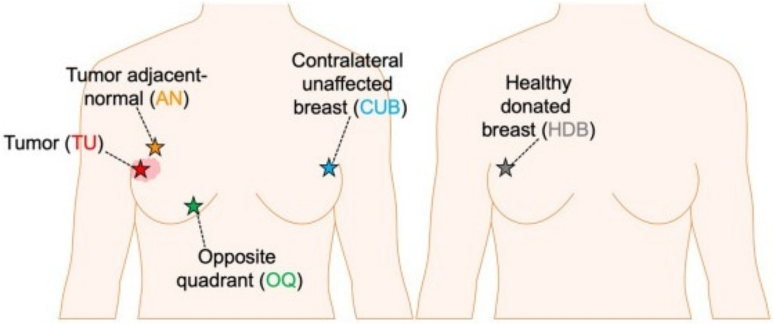


Figure 3.11: Schematic representation of the five tissue categories collected along the tumor proximity axis (TPxA), reproduced from [24].

For cross-dataset consistency, only three biologically aligned classes are retained for downstream analyses: HDB (treated as Normal baseline), AN (Adjacent), and TU (Tumor). OQ and CUB samples are excluded, as they represent intermediate benign tissues along the proximity axis and are not directly comparable with the three-class framework adopted in the other cohorts.

Global methylation distributions The group-wise mean β-value density curves (Figure 3.12) display the expected bimodal configuration of Illumina methylation arrays. A dominant peak is observed at high methylation levels, accompanied by a smaller peak consistent with standard EPIC methylation profiles [25].

Although the overall shapes remain comparable across tissue groups, a clear ordering is visible at the high-β peak. Normal samples exhibit the highest density, Adjacent samples show a slightly attenuated peak, and Tumor samples present the lowest amplitude together with a broader distribution extending toward intermediate β values. This progressive reduction in the fully methylated peak is consistent with increasing methylation variability along the Normal–Adjacent–Tumor axis.

CpG-level instability and recurrent outliers The heatmap of the top outlier CpG loci (Figure 3.13a) shows that epigenetic instability is not uniformly distributed. Alterations concentrate at specific CpG sites and within specific samples, forming vertical and horizontal bands rather than a diffuse genome-wide pattern. Both directions of deviation are observed, with focal hypermethylation and focal hypomethylation present within the selected loci [10].

The corresponding $\Delta\beta$ distributions (Figure 3.13b) highlight distinct profiles for the two contrasts. The Tumor–Normal curve is broader and displays a higher density of small negative shifts, indicating widespread but moderate hypomethylation in Tumor samples. In contrast, the Adjacent–Normal distribution is more sharply centered around zero but exhibits a comparatively more pronounced negative tail,

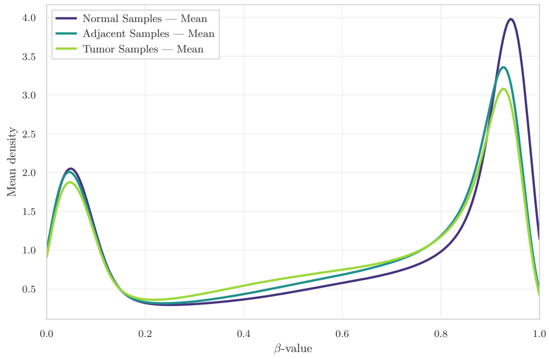
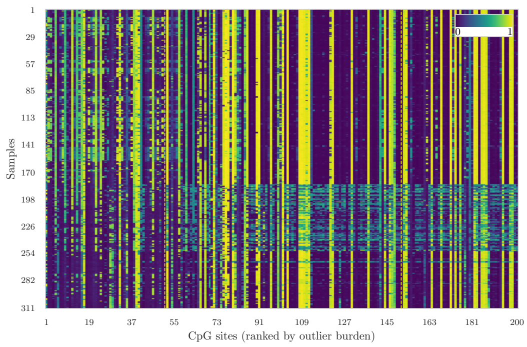
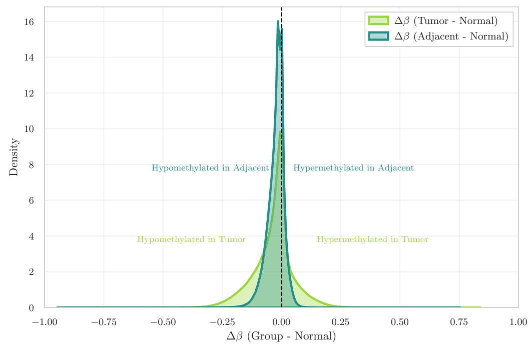


Figure 3.12: Group-wise mean β-value density in GSE287331.



(a) Top outlier CpG loci.



(b) $\Delta\beta$ distributions (Tumor/Adjacent − Normal).

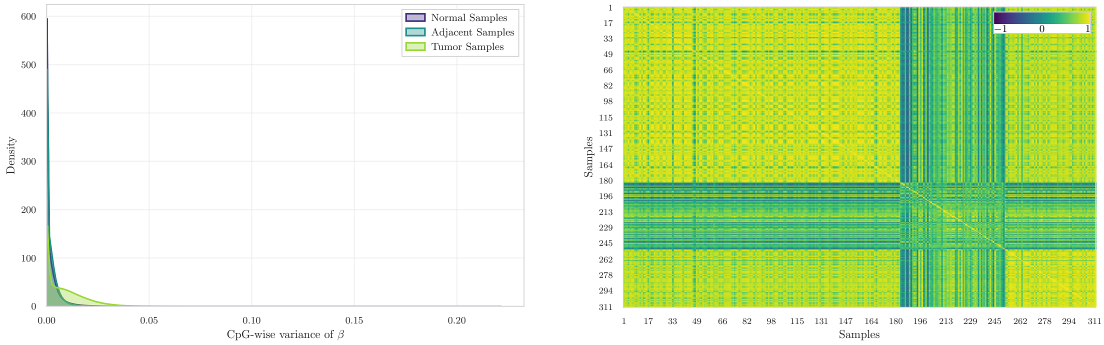
Figure 3.13: Heatmap of the top outlier CpGs (left) and corresponding $\Delta\beta$ distributions (right), illustrating CpG-level instability in GSE287331.

reflecting fewer yet larger hypomethylated deviations. Overall, hypomethylation represents the dominant direction of change in both comparisons.

Variance and correlation structure The CpG-wise variance density curves (Figure 3.14a) show that Normal and Adjacent samples are nearly superimposed, with both groups exhibiting extremely low and tightly concentrated variance values. Tumor samples display a slightly broader right tail, indicating modestly increased genome-wide variability; however, the overall distribution remains sharply peaked near zero.

At the level of CpG-wise variance, Adjacent tissue remains closely aligned with the Normal baseline, whereas Tumor samples show only a mild elevation in variability.

A complementary perspective is provided by the sample correlation heatmap



(a) CpG-wise variance distributions.

(b) Sample correlation heatmap.

Figure 3.14: Variance and correlation structure across Normal, Adjacent, and Tumor samples in GSE287331.

(Figure 3.14b). Correlations are uniformly high across the cohort, consistent with the predominance of stable CpG loci in genome-wide methylation data [7]. Although tissue labels are not overlaid on the matrix, localized blocks of higher similarity are visible, indicating structured relationships within subsets of samples rather than abrupt global separation.

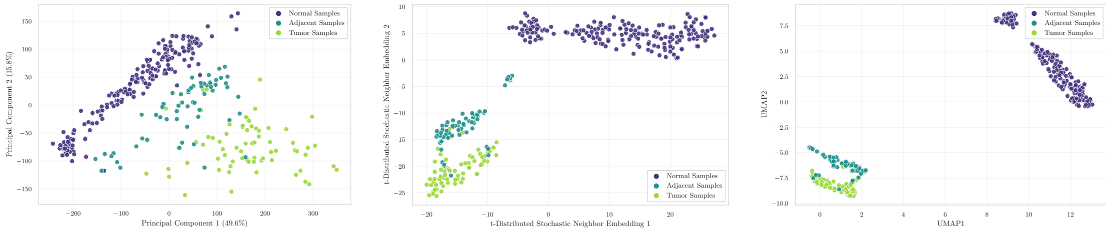
Low-dimensional embeddings Dimensionality reduction using PCA, t-SNE, and UMAP was applied to visualise the global organisation of methylation profiles (Figure 3.15).

Across all three embeddings, a consistent and well-defined structure is observed. Normal and Adjacent samples form two clearly separated clusters. The separation is visible in PCA space, becomes more pronounced in t-SNE, and appears as fully detached manifolds in the UMAP projection.

Tumor samples occupy a distinct and more dispersed region of the embedding space, reflecting increased epigenetic heterogeneity. Trustworthiness scores are uniformly high (0.961–0.980), indicating excellent preservation of local neighbourhood structure in all low-dimensional projections. No substantial overlap between Tumor and the other two groups is observed in any of the projections, indicating strong intrinsic group structure in this cohort.

Genome annotation coverage The genomic distribution of annotated CpG contexts is dominated by Open Sea regions, followed by CpG Islands, then Shores, with Shelves representing the smallest fractions.

This ordering is consistent with the established architecture of EPIC arrays [18, 19].

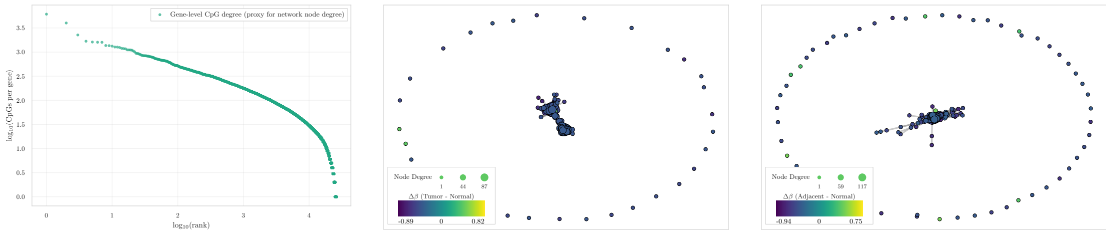


(a) PCA trustworthiness = 0.961.

(b) t-SNE trustworthiness = 0.980.

(c) UMAP trustworthiness = 0.973.

Figure 3.15: Low-dimensional embeddings of GSE287331 samples.



(a) Gene-level CpG degree distribution.

(b) DNB-like network: Tumor vs Normal.

(c) DNB-like network: Adjacent vs Normal.

Figure 3.16: Dynamic-network proxy visualizations for GSE287331.

Dynamic-network proxies The degree–rank curve (Figure 3.16a) shows a heavy-tailed distribution of CpG counts per gene, consistent with a hub–periphery architecture typical of biological networks.

Both $\Delta\beta$-based network layouts — Tumor vs Normal and Adjacent vs Normal (Figure 3.16b, Figure 3.16c) — share a similar overall structure: a dense and compact central block of nodes connected by short-range edges, surrounded by a ring of peripheral, weakly connected nodes. This geometry reflects the underlying degree distribution and indicates that most CpG–gene units contribute only minimally to group-level differences.

Despite this shared organisation, the Tumor vs Normal comparison exhibits a more pronounced central cluster with larger $\Delta\beta$ magnitudes, suggesting that a restricted subset of CpG–gene units undergoes locally coordinated perturbations. In contrast, the Adjacent vs Normal layout preserves the same global shape but shows weaker deviations, indicating only mild coordination of methylation changes.

Overall interpretation The exploratory analysis of GSE287331 indicates that, at the whole-methylome level, the three tissue states retain a broadly preserved global methylation architecture. Group-wise mean β-value densities follow the typical bimodal EPIC profile with only a subtle gradient (Normal > Adjacent > Tumor), and both CpG-wise variance distributions and sample correlation structure show high overall similarity across tissues, with only mild variability increases in Tumor samples.

In contrast to the previous cohorts, low-dimensional embeddings reveal a clear and consistent separation of all three groups. Normal and Adjacent samples form distinct clusters in PCA, with the separation further strengthened in t-SNE and UMAP, where the groups appear as detached manifolds. Tumor samples occupy a separate and more diffuse region, reflecting increased epigenetic heterogeneity. These projections indicate that, despite similar first- and second-order global summaries, multivariate structure in this dataset captures strong group-level organisation.

Locus-specific analyses reveal focused disruptions. Tumor samples exhibit widespread mild hypomethylation, whereas Adjacent tissue shows fewer but larger-magnitude deviations. Dynamic-network proxies further indicate a compact and perturbed central core in the Tumor comparison, while Adjacent–Normal differences display weaker coordination, consistent with early-stage methylation alterations.

Overall, GSE287331 emerges as a structurally coherent and biologically informative dataset in which Normal–Adjacent–Tumor separation is markedly more pronounced than in the other cohorts, while global methylation architecture remains broadly conserved.

3.4 Inter Dataset Exploration and Comparison

The objective of this section is not to merge the datasets—an approach that would be biologically and technically inappropriate due to differences in array platforms (HM450 vs. EPIC), preprocessing pipelines—but rather to compare their internal behaviours. Such a cross-dataset perspective enables the assessment of whether the early epigenetic alterations identified in the intra-dataset analyses are *reproducible across independent cohorts*, thereby providing support for the central biological premise of this thesis: the detection of early methylation drift in histologically normal tissues and the evaluation of its potential contribution to tumour initiation.

3.4.1 Analytical Framework

Across all three datasets, the intra-dataset analyses revealed three highly consistent phenomena:

- Normal and Adjacent tissues are globally similar, with nearly superimposed β-value distributions and comparable CpG-wise variance profiles.

Table 3.4: Cross-dataset comparison of Normal (N) vs Adjacent (A) contrasts using global and instability metrics.

Dataset	Mean $\lvert\Delta\beta\rvert$ (A–N)	Var Ratio (A/N)	Outlier Ratio (A/N)	Silhouette (A/N)
GSE69914	0.009	0.988	1.353	0.003
GSE225845	0.015	0.972	2.790	0.042
GSE287331	0.031	0.930	13.329	0.344

- Tumour samples show increased heterogeneity and a clear bias toward hypomethylation, although the magnitude of this shift varies across datasets.

- Adjacent tissues exhibit early methylation drift, detectable through $\Delta\beta$ distributions and outlier profiles even when global metrics remain highly similar to Normal.

These patterns are consistent with the expected biological progression from Normal to Tumour, in which epigenetic deregulation emerges gradually and affects normal-appearing tissue surrounding the tumour. The inter-dataset analysis therefore evaluates the *directional consistency* of these effects rather than aiming at strict quantitative comparability.

3.4.2 Cross-Dataset Comparative Metric

To quantify the epigenetic deviation of Adjacent tissue from the Normal methylome across the three cohorts, a set of global and instability-oriented metrics was computed independently within each dataset for the Adjacent versus Normal contrast.

All metrics reported in Table 3.4 were computed using the complete CpG set available in each dataset, rather than the cross-platform intersection. The resulting values therefore reflect cohort-specific dimensionality and internal variance structure.

Collectively, these metrics indicate the presence of early epigenetic drift in histologically non-tumour tissue. Across cohorts, increasing mean $|\Delta\beta|$, progressive variance imbalance, elevated outlier ratios, and improved silhouette separation consistently support a directional shift from Normal to Adjacent states, in agreement with the field-defect framework described by Teschendorff et [12] and reinforced in other tissue contexts.

Mean absolute methylation difference For each CpG, the absolute mean difference between Adjacent and Normal was defined as

$$|\Delta\beta_i| = \left|\beta_i^{(A)} - \beta_i^{(N)}\right|. \tag{3.2}$$

The cross-cohort pattern reveals progressively larger locus-specific shifts from GSE69914 to GSE225845 and GSE287331. Although the absolute magnitude of the effect remains modest, it is consistent in direction across cohorts. This monotonic increase reflects a strengthening early deviation in Adjacent tissues. Comparable locus-wise methylation drifts in histologically normal tissue have been reported in independent systems, such as colonic mucosa adjacent to adenomas [26], supporting the interpretation of these alterations as early manifestations of field cancerization.

Global variance ratio For each sample s, the global methylation variance was computed across all CpG sites: $v_s = \mathrm{Var}(\beta_{s,1}, \beta_{s,2}, \ldots, \beta_{s,m})$, where m denotes the number of CpGs. Group-wise mean variances were then defined as

$$\overline{v}^{(N)} = \frac{1}{|N|}\sum_{s\in N} v_s, \qquad \overline{v}^{(A)} = \frac{1}{|A|}\sum_{s\in A} v_s, \tag{3.3}$$

and the reported metric corresponds to their ratio:

$$\frac{\mathrm{Var(A)}}{\mathrm{Var(N)}} = \frac{\overline{v}^{(A)}}{\overline{v}^{(N)}}. \tag{3.4}$$

Across cohorts, this ratio remained consistently close to unity, indicating that Adjacent tissue does not exhibit diffuse genome-wide instability. Such behaviour suggests that early pre-neoplastic alterations are predominantly *focal* rather than global, with only specific CpG sites displaying abnormal dispersion while overall variance remains largely preserved [26].

This interpretation is consistent with contemporary models of early tumourigenesis, in which subtle and locally restricted perturbations accumulate prior to large-scale chromatin deregulation and overt neoplastic transformation [27].

Outlier burden ratio For each sample, the outlier burden is defined as the number of CpG sites whose methylation deviates from the Normal reference distribution by more than three times the Normal interquartile range:

$$\mathrm{outlier}_s = \#\left\{ i \ : \ \left|\beta_{i,s} - \tilde{\beta}_i^{(N)}\right| > 3 \cdot IQR_i^{(N)} \right\}, \tag{3.5}$$

where $\tilde{\beta}_i^{(N)}$ and $IQR_i^{(N)}$ denote the median and interquartile range of CpG i across Normal samples.

The reported A/N ratio corresponds to the mean outlier burden in Adjacent tissue divided by the mean burden in Normal tissue. Across cohorts, this metric exhibits the most pronounced cross-dataset gradient, with progressively stronger enrichment of outlier events in Adjacent tissue. Such behaviour mirrors evidence that increased local methylation variability in histologically normal tissue represents a sensitive marker of early instability [26], and provides quantitative support for the presence of marked field effects in GSE287331.

Silhouette score For each sample s, let $a(s)$ denote the mean distance from s to all other samples within the same group (cohesion), and let $b(s)$ denote the smallest mean distance from s to samples in the alternative group (separation). The silhouette coefficient for sample s is defined as

$$\mathrm{sil}(s) = \frac{b(s) - a(s)}{\max\{a(s),\, b(s)\}}, \tag{3.6}$$

and the reported score corresponds to the mean silhouette computed across all Normal and Adjacent samples.

Higher values indicate clearer multivariate separation between Normal and Adjacent states, whereas values near zero reflect substantial overlap. Across cohorts, the silhouette analysis reveals limited separation in GSE69914 and GSE225845, consistent with minimal global displacement in multivariate space. In contrast, GSE287331 exhibits partial separation, indicating that the epigenetic landscape of Adjacent tissue is measurably shifted relative to Normal. This behaviour aligns with mechanistic models in which epigenetic dysregulation constitutes one of the earliest events in tumour initiation [27], preceding morphological transformation.

Collectively, the set of metrics delineates a consistent Normal–Adjacent gradient across datasets. Early epigenetic drift is detectable in all cohorts, while its magnitude, focality, and multivariate visibility progressively intensify from GSE69914 to GSE287331. Importantly, the combination of stable global variance with pronounced outlier enrichment supports a model in which early pre-neoplastic changes arise through *localised* perturbations, a hallmark of field cancerization [26, 12].

3.4.3 Visual Comparative Analysis

To complement the quantitative summary reported in Table 3.4, this section provides a visual characterisation of the Normal–Adjacent contrast across the three cohorts. All analyses are performed on the three-way intersection of 326,330 CpG sites, ensuring maximal comparability across platforms (HM450 and EPIC) and preprocessing pipelines [18]. The objective is not to achieve numerical concordance at the single-CpG level—a notoriously difficult task due to platform differences, probe-design effects, and cohort-specific processing [19, 28]—but rather to evaluate whether the *shape*, *direction*, and *global behaviour* of early methylation drift remain coherent across studies.

Global $\Delta\beta$ distributions The overlaid $\Delta\beta$ (Adjacent–Normal) density curves across the three cohorts are shown in Figure 3.17. All distributions are sharply centred around zero, indicating that the N–A methylation drift remains globally subtle in each dataset, consistent with the focal and low-amplitude nature of early

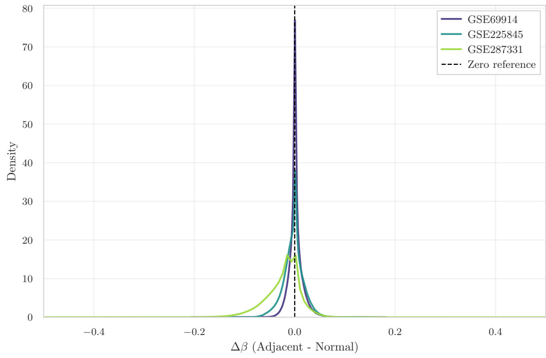


Figure 3.17: Overlay of $\Delta\beta$ (Adjacent–Normal) density curves across the three datasets, computed on the 326,330 CpG sites shared by all cohorts.

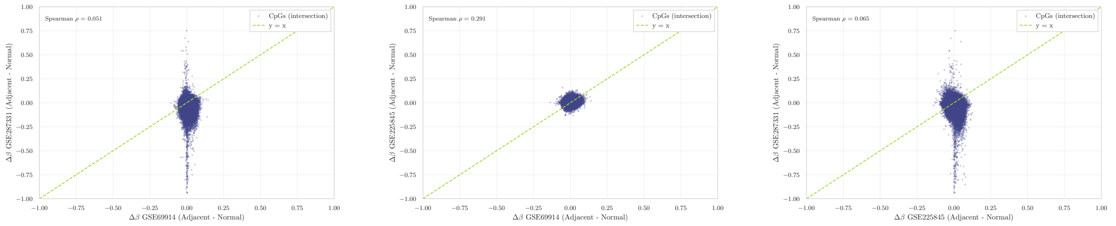
epigenetic alterations described in pre-neoplastic tissues [12, 29]. Despite differences in dispersion—reflecting platform-dependent noise, probe-type composition, and study-specific variability [16, 19]—the three curves exhibit a highly similar shape. This pattern indicates that early epigenetic deviations in Adjacent tissue, although small in magnitude, exhibit reproducible global distributional features across independent cohorts, rather than exact concordance at the single-CpG level.

Pairwise $\Delta\beta$ concordance across datasets To assess whether locus-specific A–N shifts replicate across cohorts, pairwise scatterplots are shown for all dataset pairs (Figure 3.18a, Figure 3.18b, Figure 3.18c). Each point corresponds to a CpG in the three-way intersection; axes represent the mean $\Delta\beta$ (A–N) computed independently within each dataset. Across all pairs, Spearman correlations remain weak ($\rho = 0.05$–0.29), consistent with the well-established difficulty of achieving single-CpG reproducibility across independent methylation studies [18, 28, 30].

The comparison between GSE69914 and GSE225845 exhibits mild monotonic agreement, whereas pairs involving GSE287331 show near-zero correlation. These results indicate that while the *global* N–A drift remains consistent across cohorts (as demonstrated by the density curves), *fine-scale, CpG-specific* effects are strongly influenced by platform architecture, batch structure, and study-specific variability [19, 28].

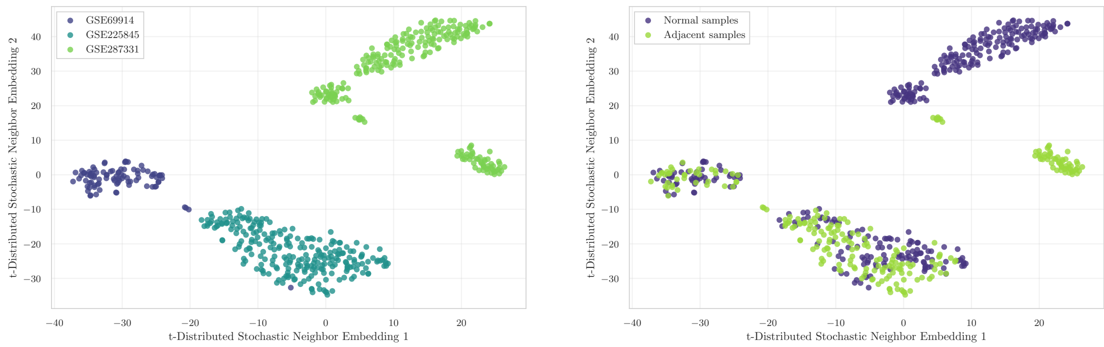
t-SNE embeddings on the shared CpG intersection To investigate the multivariate structure of the N–A contrast across cohorts, t-SNE embeddings are computed on M-values restricted to the shared CpGs. Two complementary visualisations of the embedding are provided in Figure 3.19a (coloured by dataset) and Figure 3.19b (coloured by tissue state).

When coloured by dataset, the embedding exhibits clear clustering by study,



(a) GSE69914-GSE287331, Spearman $\rho \approx 0.051$ **(b)** GSE69914-GSE225845, Spearman $\rho \approx 0.291$ **(c)** GSE225845-GSE287331, Spearman $\rho \approx 0.007$

Figure 3.18: Pairwise $\Delta\beta$ (Adjacent–Normal) scatterplots across the three methylation datasets, computed on the 326,330 shared CpG sites. Spearman ρ values are reported in each panel.



(a) t-SNE coloured by dataset trustworthiness = 0.976 **(b)** t-SNE coloured by tissue state trustworthiness = 0.976

Figure 3.19: t-SNE embeddings computed on the 326,330 CpGs shared across GSE69914, GSE225845, and GSE287331.

reflecting the dominant influence of batch and platform effects in high-dimensional methylation data [28, 30]. This pattern indicates that the global methylation structure is largely driven by inter-cohort variability, supporting the decision not to merge datasets at the raw feature level.

When coloured by tissue state (Normal vs. Adjacent), no global separation across cohorts emerges, as the embedding remains primarily structured by dataset identity. However, within individual study-specific clusters, a state-dependent displacement becomes visible. This separation is particularly evident in GSE287331, where Normal and Adjacent samples form more clearly distinct substructures. This behaviour is consistent with early, localised methylation drift, mirroring the quantitative metrics reported in Table 3.4.

Overall, the visual analyses corroborate the quantitative metrics: early epigenetic drift is detectable in all cohorts, but its magnitude and visibility vary substantially, with GSE287331 exhibiting the clearest field-defect signature. The combination of globally subtle yet directionally consistent $\Delta\beta$ patterns, weak CpG-level concordance, and dataset-specific multivariate structure reinforces the view that the Normal→Adjacent transition reflects genuine early epigenetic deregulation rather than dataset-specific artefacts.

3.5 Synthesis of Exploratory Findings and Pre-Processing Rationale

The exploratory analyses presented in this chapter reveal a consistent picture across all three cohorts: the Normal–Adjacent transition does not manifest as a global reorganisation of the methylome, but as a subtle, focal drift that becomes detectable only through locus-specific perturbations and multivariate displacements. This characteristic makes it particularly sensitive to technical sources of variability — probe-design bias, batch effects, and platform differences — which operate at a comparable or larger scale than the biological signal of interest.

This sensitivity has direct methodological consequences. Raw methylation data, as distributed via GEO, cannot be used directly for downstream modelling without first addressing these technical confounders: doing so would risk attributing platform-specific artefacts to genuine epigenetic alterations, or conversely, suppressing the subtle Normal–Adjacent differences under overly aggressive normalisation. The challenge is therefore not simply to clean the data, but to do so in a way that preserves the biological structure identified here.

Chapter **??** addresses this challenge by developing a structured, platform-aware preprocessing pipeline. Each step is motivated by the specific limitations identified in the present chapter.

Chapter 4

Data preprocessing

4.1 Preprocessing Rationale and Technical Constraints

DNA methylation arrays provide high-dimensional, genome-wide measurements in which biological variability and platform-specific technical artefacts are inevitably superimposed. The exploratory analyses of Chapter 3 revealed a consistent picture across all three cohorts: Normal and Adjacent tissues exhibit nearly identical global methylation distributions, overlapping variance profiles, and indistinguishable sample-level correlation structure. Meaningful differences between tissue states emerge exclusively at focal CpG loci—as quantified by locus-specific $\Delta\beta$ distributions and outlier burden ratios—and remain subtle even at that scale (mean $|\Delta\beta|$ ranging from 0.009 in GSE69914 to 0.031 in GSE287331). This characteristic makes the Normal–Adjacent signal particularly vulnerable to technical confounders, which operate at a comparable or larger scale than the biological effect of interest. Preprocessing must therefore be precise: aggressive enough to remove technical noise, but conservative enough to preserve the subtle epigenetic drift that motivates this thesis.

Three main classes of technical issues must be systematically addressed before any statistical modelling can proceed. First, **probe-design bias** intrinsic to the Infinium platform arises from the coexistence of two chemically distinct probe types (Type I and Type II), which produce inherently different β-value distributions and can distort cross-sample comparability if not corrected [31, 32]. This issue is not uniform across datasets: as will be shown in Section 4.3, GSE69914 and GSE225845 were released with normalization already applied by the original authors, whereas GSE287331 shows clear diagnostic evidence of missing probe-type correction, requiring an additional preprocessing decision with non-trivial consequences for downstream group separation. Second, a subset of **unreliable or biologically**

confounded probes—including cross-reactive loci, SNP-affected sites, non-CpG targets, and sex-chromosome probes—introduces systematic measurement error that can generate spurious associations or mask genuine methylation differences [33, 34]. Third, the **statistical scale of β-values** poses inferential challenges: the bounded [0,1] support and strong mean-dependent heteroscedasticity of β-values violate the assumptions of standard parametric models, motivating transformation to a more statistically well-behaved representation prior to modelling [31].

This chapter addresses these three issues through a structured, platform-aware preprocessing pipeline applied consistently across all cohorts. Section 4.2 describes the general framework and the rationale for each step. Section 4.3 documents the dataset-specific application, focusing on the decisions that deviate from the standard workflow—most notably the handling of probe-type bias in GSE287331, where the tension between technical correction and preservation of biological structure requires explicit justification. Section 4.4 summarises the effect of preprocessing on CpG dimensionality and cross-dataset overlap, providing the basis for the feature selection strategy developed in Chapter 5.

4.2 General Preprocessing Framework

This section describes the general preprocessing framework applied to all three methylation datasets. The pipeline comprises four sequential steps, each addressing a distinct class of technical issue identified in the overview above. Two of these steps—data integrity verification and Infinium probe-design bias correction—require dataset-specific decisions whose rationale and consequences are discussed in Section 4.3. The remaining two steps—technical probe filtering and β-to-M transformation—are applied uniformly across all cohorts following the same protocol.

4.2.1 Structural Integrity and Cohort Harmonization

Prior to any preprocessing operation, each dataset undergoes a structural verification stage to ensure the consistency and completeness of the input data. This step checks for correct alignment between the methylation matrix and its associated phenotype table, verifies sample identifier uniqueness, and confirms the absence of missing metadata fields. Only samples for which both a complete methylation profile and a valid tissue label are available are retained for downstream analysis.

Probe-level and sample-level missingness in the β-value matrix is also assessed at this stage. For two of the three datasets (GSE69914 and GSE225845), the released matrices are fully observed and require no imputation. For GSE287331, a non-negligible fraction of CpG sites exhibits missing values across samples, necessitating a dedicated filtering and imputation strategy consistent with current recommendations for EPIC-array quality control [35, 36]. The dataset-specific treatment of missingness is detailed in Section 4.3.3.

Finally, cohort harmonization is applied where necessary to align tissue labels to the three-class framework (Normal, Adjacent, Tumor) adopted uniformly across datasets. Samples belonging to tissue categories outside this framework are excluded at this stage to ensure cross-dataset comparability.

4.2.2 Probe-Type Bias Diagnostics and Correction

A well-documented source of systematic bias in Illumina Infinium methylation arrays is the coexistence of two chemically distinct probe designs. Type I probes use two bead types per CpG (one for the methylated channel, one for the unmethylated channel), whereas Type II probes use a single bead type and rely on single-base extension with two-color detection. This design difference produces inherently distinct β-value distributions for the two probe types: Type I probes tend to yield more extreme values near 0 and 1, while Type II probes exhibit a broader, intermediate distribution. If uncorrected, this imbalance propagates into downstream statistical analyses and can generate probe-type-specific artefacts in differential methylation testing [7, 16].

The canonical approach to mitigating this bias is Beta-Mixture Quantile normalisation (BMIQ), introduced by Teschendorff et al. [7]. BMIQ explicitly models probe-type bias at the β-value level through a state-aware distributional alignment procedure. For each sample independently, a three-component Beta mixture model is first fitted to the β-value distribution of Type I probes, representing unmethylated, partially methylated, and fully methylated states. These fitted components define the reference state-specific distributions. Type II probes are then probabilistically assigned to one of the three methylation states based on their β-values, and within each state a quantile-mapping step is applied to align the Type II distribution to the corresponding Type I reference distribution. Because the procedure is performed independently for each sample, BMIQ does not introduce cross-sample information or global harmonisation across tissue classes; it strictly corrects within-sample probe-type bias.

For each dataset, a systematic diagnostic is performed prior to any correction step [7, 37]: probes are stratified by design type using the official Illumina manifest, and two complementary assessments are carried out—(i) a density comparison of β-value distributions for Type I and Type II probes, and (ii) a quantile–quantile plot of Type II versus Type I quantiles, with the median absolute deviation (MAD) of the empirical Q–Q curve from the identity line as a scalar diagnostic metric. Datasets for which the original authors applied BMIQ or an equivalent normalisation are expected to exhibit near-overlapping distributions and a MAD close to zero; datasets lacking probe-type correction will display a pronounced distributional mismatch and elevated MAD.

The outcome of this diagnostic, and the corrective action taken, differs across cohorts and is reported in full in Section 4.3. Critically, the decision of whether to apply BMIQ is not made mechanically: as discussed in Section 4.3.3, BMIQ correction can attenuate genuine biological structure when group-level differences are expressed as broad distributional shifts, a phenomenon documented in systematic evaluations of EPIC-array normalisation methods [1, 38, 39]. The choice between correcting and preserving the raw data is therefore made on the basis of empirical evidence from the diagnostic plots and quantitative assessment of group separability before and after correction.

4.2.3 Technical Probe Filtering

After addressing probe-design bias, unreliable CpG loci are removed through a systematic filtering procedure based on curated external annotation resources. The objective is to exclude probes whose measured β-values may reflect technical measurement error rather than genuine DNA methylation, thereby reducing noise and improving the reliability of downstream differential analyses. Four categories of technically problematic probes are systematically excluded.

Cross-reactive and multi-mapping probes. Probes that hybridise to multiple genomic loci produce β-values that cannot be unambiguously attributed to a single CpG site. These are identified using the catalogues of Naeem et al. [40], Chen et al. [41], Pidsley et al. [19], Zhou et al. [18], and McCartney et al. [42]. Full descriptions of each resource are provided in Appendix A.

SNP-affected probes. Probes whose interrogated CpG dinucleotide, single-base extension site, or probe body overlaps a common genetic variant may measure genotype rather than methylation status, producing population-stratified artefacts in group comparisons [18, 19]. These are excluded using the `MASK_snp5` and `MASK_extBase` fields from the Zhou annotation [18], supplemented by variant-aware lists from Naeem et al. [40].

Non-CpG probes. Probes targeting non-CpG cytosine contexts (identified by the `ch` prefix in Illumina probe identifiers) are excluded, as their interpretation differs from canonical CpG methylation and is outside the scope of this analysis [18, 19].

Sex-chromosome probes. Probes located on chromosomes X and Y are removed to prevent sex-driven confounding effects. Sex-chromosome CpGs exhibit substantially different methylation profiles between males and females, and including them in joint normalisation has been shown to introduce artificial sex-related bias into autosomal probes [43]. Since the datasets analysed here contain samples from both sexes and sex-specific methylation is not the object of investigation, removal therefore represents a conservative strategy to prevent confounding without compromising autosomal inference [16].

The filtering resources are applied under a union criterion: a probe is removed if it appears in at least one curated list. Platform-specific manifests (HumanMethylation450 v1.2 for GSE69914 [44]; MethylationEPIC v1.0 B5 for GSE225845 and GSE287331 [45]) are used for manifest-based validation, chromosome annotation, and non-CpG probe identification. The number of probes flagged by each resource, and the redundancy of flags across resources, are reported per-dataset in Section 4.3.

4.2.4 Statistical Transformation: β-to-M Values

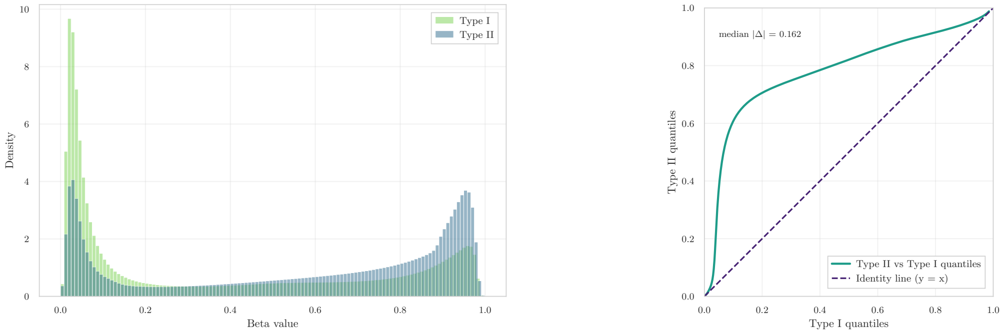
The final preprocessing step transforms the filtered β-value matrix into M-values prior to statistical modelling. As defined in Equation (3.1), β-values are bounded proportions in [0,1] derived from methylated and unmethylated probe intensities. While β-values are intuitive and directly interpretable as fractional methylation, their bounded support induces strong mean-dependent heteroscedasticity: variance is inflated near intermediate values ($\beta \approx 0.5$) and compressed near the boundaries, violating the homoscedasticity assumptions underlying standard linear modelling frameworks [6].

The logit transformation to M-values,

$$M = \log_2\left(\frac{\beta + \varepsilon}{1 - \beta + \varepsilon}\right), \quad \varepsilon = 10^{-6}, \tag{4.1}$$

maps $\beta \in (0,1)$ to an unbounded real-valued scale and substantially stabilises the mean–variance relationship [6]. As shown in their analysis, the relationship between β and M-values is a logit transformation yielding an approximately linear correspondence in the intermediate methylation range and nonlinear compression at the extremes. The small offset ε is introduced to handle boundary values numerically. The resulting M-value distributions are approximately symmetric and exhibit markedly flatter variance profiles across the full methylation range, making them the standard input for differential methylation analysis within linear modelling frameworks and related inferential procedures [31, 32, 46].

It is important to note that M-values are adopted exclusively for statistical modelling: β-values are retained in parallel for biological interpretation, effect-size quantification, and visualisation, following the dual-representation convention recommended by Du et al. [6] and widely adopted in the literature [46, 47].



(a) β-value density by probe type. **(b)** Q–Q plot (Type II vs. Type I).

Figure 4.1: Infinium Type I/II bias diagnostics in GSE69914.

The transformation is applied to the complete post-filtering matrix for each dataset. Distributional diagnostics confirming the expected variance-stabilising effect are reported in Section 4.3. This choice is consistent with the feature-selection and modelling procedures developed in the subsequent chapter **??**, which rely on variance-based ranking, linear statistics, and stability criteria that assume approximately homoscedastic behaviour.

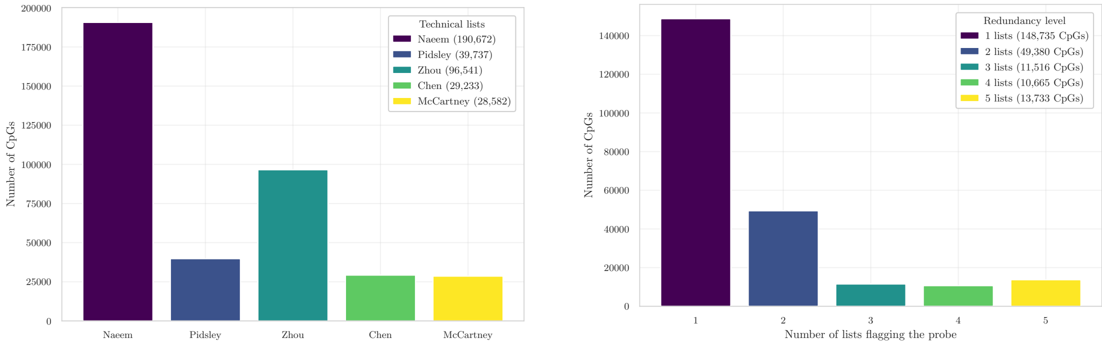
4.3 Dataset-Specific Implementation

This section documents the dataset-specific application of the preprocessing pipeline described in Section 4.2. For each cohort, the outcome of structural verification, probe-type bias diagnostics, technical probe filtering, and β-to-M transformation is reported in detail. Although the general framework is applied consistently across all datasets, certain steps—most notably probe-type bias correction and handling of missing values—require cohort-specific decisions. These are explicitly justified within each subsection.

The results are presented separately for GSE69914 (Section 4.3.1), GSE225845 (Section 4.3.2), and GSE287331 (Section 4.3.3).

4.3.1 Dataset GSE69914

Structural Integrity and Cohort Harmonization The dataset comprises 397 samples and 485,512 CpGs at load time. All sample identifiers are unique, and the phenotype table contains complete metadata for the required fields. Alignment between the phenotype annotation and the β-value matrix is exact. The methylation matrix is fully observed, with no missing CpG values and no incomplete samples. No exclusion or imputation was required.



(a) CpGs flagged by each technical list. **(b)** Redundancy across filtering lists.

Figure 4.2: Technical filtering summary in GSE69914.

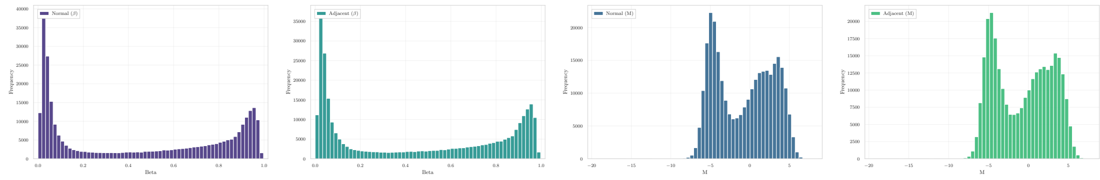
Table 4.1: CpG filtering summary for the GSE69914 dataset.

Initial CpGs	Naeem	Pidsley	Zhou	Chen	McCartney	Non-CpG	X/Y	Final CpGs
485,512	190,672	39,737	96,541	29,233	28,582	875	7,728	251,483

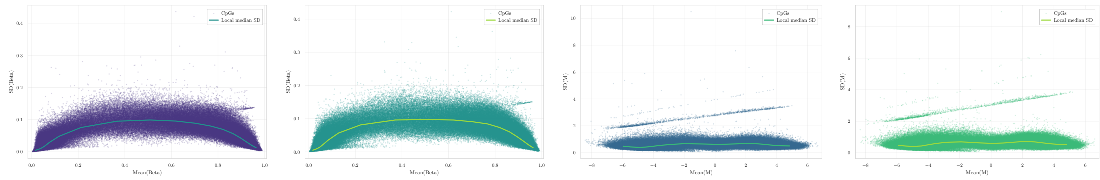
Probe-Type Bias Diagnostics and Correction Raw intensity data were reported to have been processed by the original authors using the `minfi` pipeline with BMIQ normalization [3]. Diagnostic evaluation confirms that probe-type bias is effectively mitigated in the released matrix.

The β-value density distributions of Type I and Type II probes exhibit substantial overlap, with no evidence of the characteristic multi-modal distortion observed in uncorrected data (Figure 4.1a). Quantile–quantile analysis further supports this result: the Type II versus Type I quantiles align closely to the identity line, with median absolute deviation median$|\Delta| = 0.162$, indicating balanced probe-type scaling (Figure 4.1b). Together, these diagnostics are consistent with effective BMIQ correction, in agreement with the normalization behaviour described in [7, 16, 37].

Technical Probe Filtering Intersection with external technical filtering resources resulted in substantial probe removal, as summarised in Table 4.1. The largest individual contributions derive from the Naeem and Zhou annotations, while the remaining resources flag smaller but non-negligible subsets. The redundancy structure across lists is shown in Figure 4.2b. Because these catalogues target partially distinct classes of technical artefacts—such as cross-reactivity, SNP interference, and repeat-associated probes—limited overlap across lists is expected. Most excluded CpGs are supported by a single resource, with progressively fewer loci flagged by multiple independent annotations. This pattern reflects complementary



(a) Normal (β) **(b)** Adjacent (β) **(c)** Normal (M) **(d)** Adjacent (M)



(e) Mean–SD Normal (β) **(f)** Mean–SD Adjacent (β) **(g)** Mean–SD Normal (M) **(h)** Mean–SD Adjacent (M)

Figure 4.3: Distributional comparison of β and M in Normal and Adjacent tissues in GSE69914.

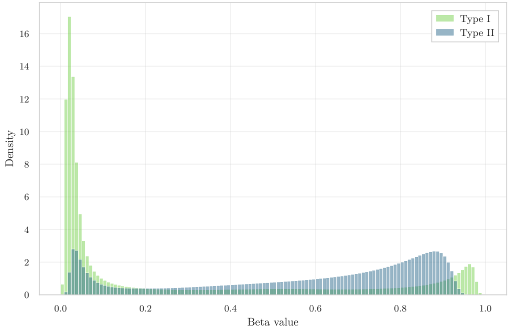
rather than redundant filtering criteria, while the subset of probes flagged by several lists indicates concordance for loci exhibiting multiple technical vulnerabilities.

Annotation-based validation using the HumanMethylation450 v1.2 manifest [48] further excluded non-CpG probes and loci lacking reliable genomic annotation. Subsequent removal of sex-chromosome probes (X/Y), consistent with established preprocessing recommendations in mixed-sex cohorts [16, 43], ensured a strictly autosomal working set.

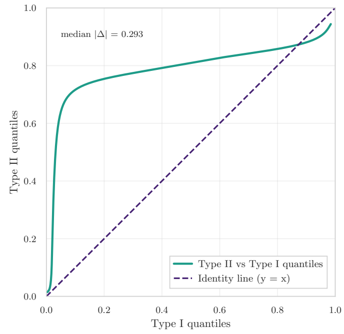
A detailed per-probe exclusion summary is available in the project repository at `removed_cpgs_all_filters_summary_gse69914.csv`.

Statistical Transformation: β-to-M Values Following transformation, the empirical distributions confirm the expected variance-stabilising behaviour of M-values (Figure 4.3). In β-space, both Normal and Adjacent tissues exhibit the characteristic bounded, U-shaped distribution with accumulation near 0 and 1. After logit transformation, the corresponding M-value distributions become substantially more symmetric and centered, while preserving the relative structure between tissue groups. The mean–SD relationships further illustrate the scale effect. In β-space, variance is strongly mean-dependent and inflates near intermediate methylation levels, producing a pronounced curvature in the mean–SD plots. In contrast, M-values display markedly flatter dispersion profiles across the full range, indicating substantial reduction of heteroscedasticity.

This behaviour is fully consistent with the theoretical and empirical findings of Du et al. [6], which motivate the use of M-values for inferential modelling.



(a) β-value density by probe type.



(b) Q–Q plot (Type II vs. Type I).

Figure 4.4: Infinium Type I/II bias diagnostics in GSE225845.

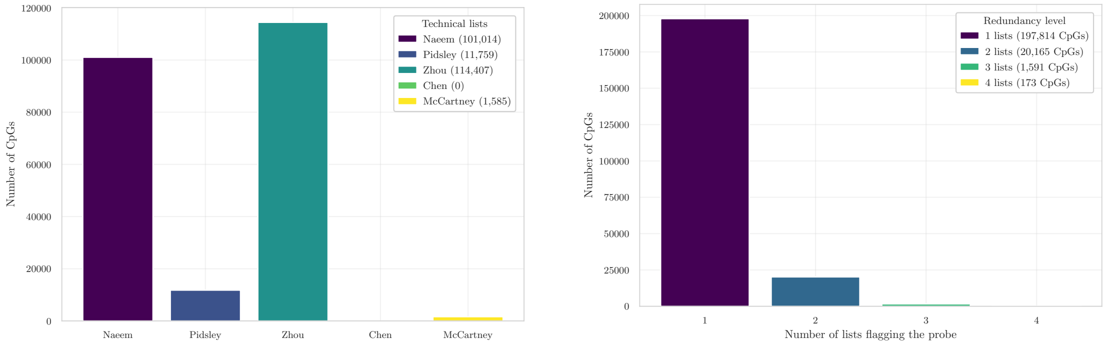
4.3.2 Dataset GSE225845

Structural Integrity and Cohort Harmonization The dataset comprised 477 samples and 750,426 CpGs at load time, consistent with the dimensionality of the Illumina EPIC platform. The phenotype table contained 477 entries, and alignment between phenotype annotation and the β-value matrix was exact. No duplicated entries were detected, and all required metadata fields were present.

Probe-level and sample-level missingness assessment confirmed that the methylation matrix was fully observed. No missing CpG values and no incomplete samples were detected. Consequently, no imputation or early NaN filtering was required prior to probe-type bias diagnostics and technical filtering.

Probe-Type Bias Diagnostics and Correction Stratification by probe design yielded 119,760 Type I and 627,842 Type II probes. Raw intensity data were reported by the original authors to have been processed using the `minfi` pipeline with BMIQ normalization (v1.4) [4]. The β-value density distributions (Figure 4.4a) show substantial overlap between probe families. Type I probes retain the characteristic concentration near the unmethylated and fully methylated boundaries, while Type II probes display a broader distribution; however, no pronounced multimodal distortion or systematic separation is observed. Quantile–quantile analysis (Figure 4.4b) further supports this interpretation. Although minor deviations from the identity line ($y = x$) are visible, the overall alignment indicates largely balanced probe-type scaling. The median absolute deviation median$|\Delta| = 0.293$ suggests that probe-type differences are attenuated but not entirely eliminated, consistent with residual variation commonly observed after BMIQ normalization [7, 16, 37].

Overall, the diagnostic patterns are compatible with prior application of probe-type bias correction.



(a) CpGs flagged by each technical list. **(b)** Redundancy across filtering lists.

Figure 4.5: Technical filtering summary in GSE225845.

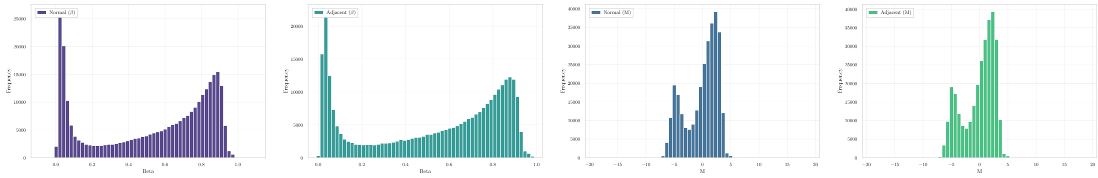
Table 4.2: CpG filtering summary for the GSE225845 dataset.

Initial CpGs	Naeem	Pidsley	Zhou	Chen	McCartney	Non-CpG	X/Y	Final CpGs
750,426	101,014	11,759	114,407	0	1,585	773	14,071	530,683

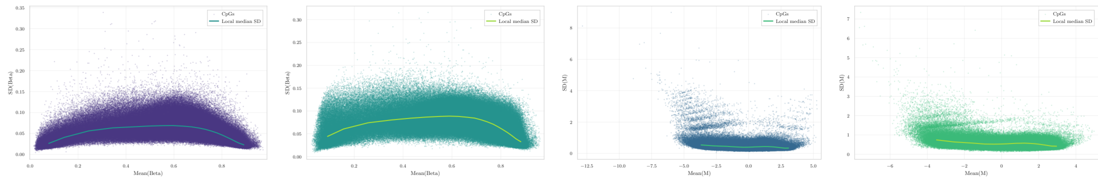
Technical Probe Filtering Intersection with external technical filtering resources resulted in substantial probe removal, as summarised in Table 4.2. The largest individual contributions derive from the Zhou and Naeem annotations, while the remaining resources flag smaller subsets. The redundancy structure across lists is shown in Figure 4.5b. A detailed per-probe exclusion summary is available in the project repository at `removed_cpgs_all_filters_summary_gse225845.csv`.

Statistical Transformation: β-to-M Values All filtered β-values were transformed to M-values using the logit relation defined in Equation (4.1). The empirical distributions (Figure 4.6) show the characteristic bounded, U-shaped profile in β-space for both Normal and Adjacent tissues. After transformation, the corresponding M-value distributions become substantially more symmetric and centered, while maintaining the relative structure between tissue classes.

The mean–SD relationships further illustrate the scale effect. In β-space, variance exhibits pronounced mean dependence, with inflation at intermediate methylation levels and compression near the boundaries. In contrast, M-values display markedly flatter dispersion profiles across the full range, indicating substantial reduction of heteroscedasticity.



(a) Normal (β) **(b)** Adjacent (β) **(c)** Normal (M) **(d)** Adjacent (M)



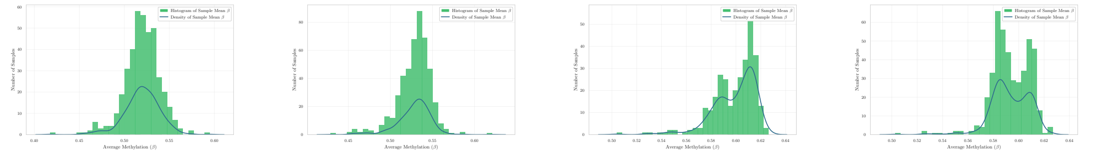
(e) Mean–SD Normal (β) **(f)** Mean–SD Adjacent (β) **(g)** Mean–SD Normal (M) **(h)** Mean–SD Adjacent (M)

Figure 4.6: Distributional comparison of β and M in Normal and Adjacent tissues in GSE225845.

4.3.3 Dataset GSE287331

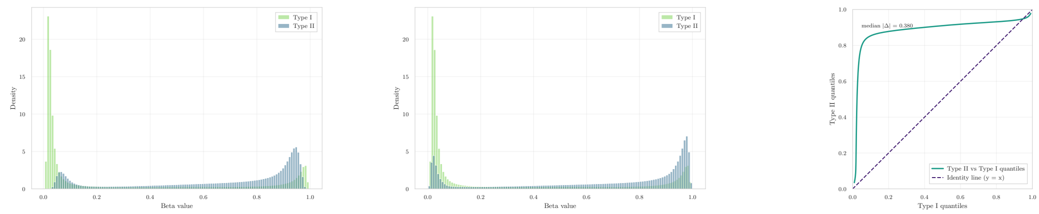
Structural Integrity and Cohort Harmonization The dataset was imported from its Parquet representation, yielding an initial β-value matrix of 446 samples and 866,552 CpGs. The phenotype table contained the same number of entries, with all required metadata fields present and alignment between phenotype annotation and the β-value matrix exact.

Unlike GSE69914 and GSE225845, the methylation matrix was not fully observed. Probe-wise missingness is a well-known source of unreliability in Illumina methylation arrays: Islam *et al.* [35] excluded probes with missing β-values in more than 2% of samples, while Mansell *et al.* [36] applied a stricter 1% threshold for detection failure in their EPIC-array quality control pipeline. Consistent with these recommendations, all CpG sites missing in more than 1% of samples were removed, prioritising high-confidence measurements while minimising reliance on imputed values and leveraging the large dimensionality of the EPIC platform. After applying this criterion, the remaining sparsity was negligible: 58,561 NaN values, corresponding to 0.0187% of all entries. Given this extremely low level of sparsity, median-based imputation is both statistically robust and distribution-preserving; residual missing values were therefore imputed using the probe-wise median β-value across samples, a strategy that avoids introducing group-specific bias and is consistent with recent harmonisation workflows such as mLiftOver, where missing values are explicitly replaced by probe-wise median substitution to preserve biologically plausible levels [49].



(a) GSE69914 **(b)** GSE225845 **(c)** GSE287331 pre BMIQ **(d)** GSE287331 post BMIQ

Figure 4.7: Per-sample mean β distributions for the three datasets.



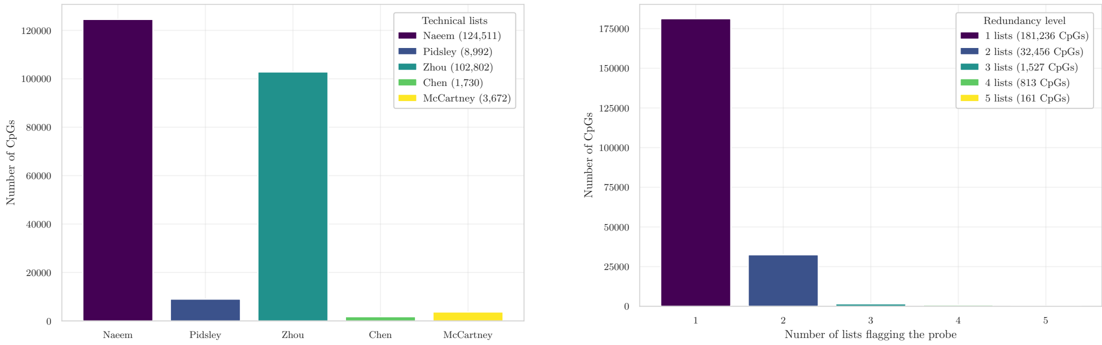
(a) GSE287331 pre BMIQ **(b)** GSE287331 post BMIQ **(c)** Q–Q comparison of Type II vs. Type I probes (pre BMIQ).

Figure 4.8: Infinium Type I/II bias diagnostics in GSE287331. Pre-BMIQ densities and Q–Q analysis indicate marked probe-type imbalance; post-BMIQ densities show partial alignment.

After harmonising phenotype labels and applying dataset-specific selection rules, only samples belonging to the three core tissue states of interest—Normal, Adjacent, and Tumour—were retained. The final working matrix consisted of 311 samples and 702,916 CpGs, with no residual missing values.

Probe-Type Bias Diagnostics and Correction Using the official EPIC v1.0 manifest, 115,705 Type I and 585,514 Type II probes were identified. Unlike GSE69914 and GSE225845, for which the original authors reported prior application of BMIQ normalisation, no probe-design bias correction was documented in the processing pipeline of GSE287331. This absence was first suggested by the per-sample mean β-value distributions (Figure 4.8): whereas GSE69914 and GSE225845 display a single broad peak, GSE287331 shows a narrower, right-shifted distribution characterised by two distinct modes in the high-β range—a pattern largely absent in the other cohorts and strongly indicative of uncorrected Type I/II probe-design bias.

Stratification of probes by design type confirmed this hypothesis. The β-value distributions of the two chemistries showed a marked mismatch (Figure 4.8a), with Type II probes displaying a strong right-shifted profile. The quantile–quantile



(a) CpGs flagged by each technical list. **(b)** Redundancy across filtering lists.

Figure 4.9: Technical filtering summary in GSE287331.

comparison confirmed a substantial systematic offset, with a median absolute deviation of 0.380 (Figure 4.8c), fully incompatible with BMIQ-corrected data. Inspection of the original publication [24] further confirmed the absence of any mention of BMIQ, SWAN, noob, or equivalent probe-type correction methods.

Given this evidence, the canonical BMIQ algorithm [7] was applied using the `wateRmelon` implementation [50], operating strictly on a per-sample basis: Type I probes define the reference mixture distribution within each sample, and Type II probes are rescaled accordingly, with no cross-sample or cross-group information introduced at any stage. Samples with insufficient numbers of probes per design type were left uncorrected to avoid unstable mixture estimation.

Post-correction diagnostics showed only a modest reduction in probe-type mismatch ($|\Delta| : 0.380 \to 0.370$), with the overall distributional discrepancy remaining substantial and the bimodality of per-sample mean β distributions persisting (Figure 4.8, Figure 4.8b). This limited harmonisation is consistent with documented caveats of quantile-based normalisation on EPIC arrays, which can artificially reduce biological variability, remove genuine group-level signal, or yield distorted distributions when differences are expressed as broad distributional shifts [38, 51, 39].

The impact on biological structure was assessed quantitatively via silhouette analysis: the average silhouette coefficient declined from $\text{sil}_{\text{pre}} = 0.2876$ to $\text{sil}_{\text{post}} = 0.2537$ after correction, indicating that BMIQ reduced the natural clusterability of the data and collapsed genuine group-level separation—particularly between Normal and Adjacent samples. For this reason, all downstream analyses are carried out on the non-BMIQ version of the dataset, which preserves sharper cluster boundaries and avoids normalisation-induced shrinkage of biological signal.

Table 4.3: CpG filtering summary for the GSE287331 dataset.

Initial CpGs	Naeem	Pidsley	Zhou	Chen	McCartney	Non-CpG	X/Y	Final CpGs
866,552	124,511	8,992	102,802	1,730	3,672	500	12,579	486,725

Technical Probe Filtering Intersection with external technical filtering resources resulted in substantial probe removal, as summarised in Table 4.3. The largest individual contributions derive from the Naeem and Zhou annotations, while the remaining resources flag smaller but non-negligible subsets (Figure 4.9a). Importantly, in the unfiltered EPIC matrix the Zhou annotation alone would have excluded 192,856 CpGs. This number substantially exceeds the net contribution observed at this stage, indicating that a considerable fraction of Zhou-flagged probes had already been removed during the earlier missingness-cleaning step. The missingness filter and the Zhou reliability masks therefore exhibit non-trivial overlap. This concordance supports the robustness of the preprocessing strategy: probes excluded due to empirical data sparsity largely coincide with loci independently annotated as technically unreliable, suggesting that early NaN-based exclusion was not arbitrary but aligned with established probe-quality criteria.

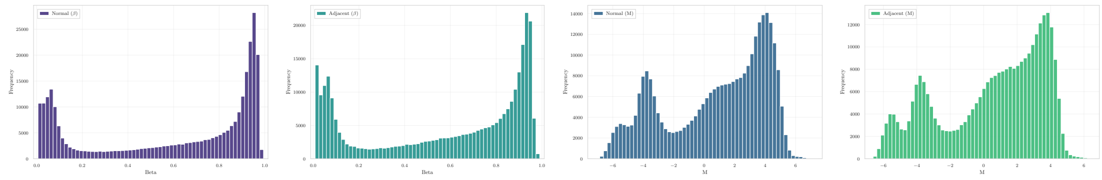
The redundancy structure across lists (Figure 4.9b) further indicates that most excluded CpGs are supported by a single resource, with progressively fewer loci flagged by multiple independent annotations. Annotation-based validation using the MethylationEPIC v1.0 manifest [45] further excluded non-CpG probes and loci lacking reliable genomic annotation. Across all filters, the dimensionality reduction is detailed in Table 4.3.

A probe-level exclusion summary is available in the project repository at `removed_cpgs_all_filters_summary_gse287331.csv`.

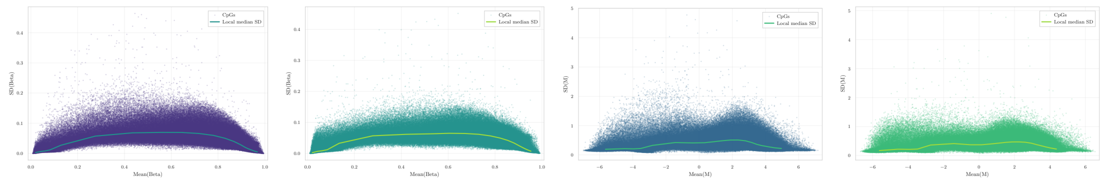
Statistical Transformation: β-to-M Values Following technical filtering, the β-value matrix was transformed into M-values using the logit relation defined in Equation (4.1), in accordance with established recommendations for variance stabilisation [6, 31, 32].

The empirical distributions (Figure 4.10) retain the characteristic bounded structure of β-values; however, in contrast to GSE69914 and GSE225845, the profiles appear less sharply bimodal and more broadly dispersed. Both Normal and Adjacent tissues exhibit reduced concentration at the methylation extremes and a wider intermediate range, consistent with the greater technical heterogeneity observed in this cohort.

After transformation, the corresponding M-value distributions become more symmetric, as expected under the logit mapping, yet remain visibly broader than in the other datasets. The mean–SD relationships confirm that while variance



(a) Normal (β) **(b)** Adjacent (β) **(c)** Normal (M) **(d)** Adjacent (M)



(e) Mean–SD (Normal, β) **(f)** Mean–SD (Adjacent, β) **(g)** Mean–SD (Normal, M) **(h)** Mean–SD (Adjacent, M)

Figure 4.10: Distributional comparison of β and M in Normal and Adjacent tissues in GSE287331.

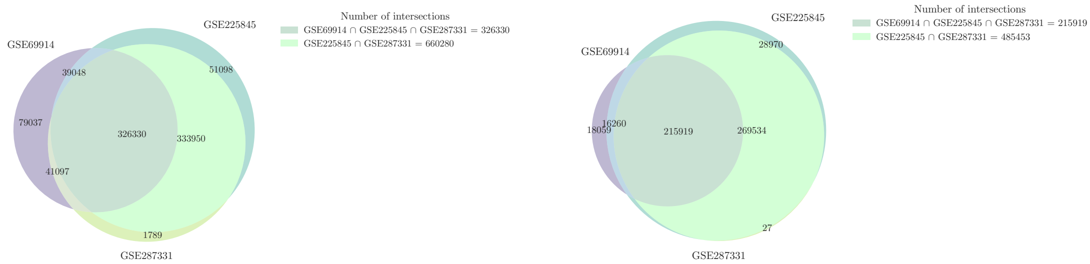
stabilisation is achieved, dispersion remains comparatively elevated across the full methylation range. This behaviour is coherent with the stronger preprocessing burden documented in earlier steps and reflects the intrinsic variability of the released matrix.

4.4 Inter-Dataset Consistency After Preprocessing

After completing the intra-dataset preprocessing for each cohort, a concise inter-dataset analysis is performed to summarise the effects of the pipeline and to motivate the subsequent feature selection step.

Consistency of the Preprocessing Framework All three datasets underwent a harmonised preprocessing workflow, including initial integrity checks, literature-based technical filtering, manifest-driven probe validation, removal of sex-chromosome loci, and transformation from β-values to M-values. While specific implementation details differ slightly depending on platform characteristics and data quality, the conceptual structure of the pipeline is consistent across datasets.

Table 4.4 reports, for each dataset, the dimensionality of the methylation matrix before and after preprocessing. Despite substantial differences in initial platform coverage (450K vs. EPIC), technical filtering consistently removes a large fraction of probes, yielding cleaner and more reliable feature spaces.



(a) CpG overlap before preprocessing.

(b) CpG overlap after preprocessing.

Figure 4.11: CpG overlap across datasets before and after preprocessing.

Table 4.4: CpG dimensionality before and after preprocessing.

Dataset	Platform	CpGs *before*	CpGs *after*
GSE69914	450K	485,512	251,483
GSE225845	EPIC	750,426	530,683
GSE287331	EPIC	866,554	486,725

CpG overlap across datasets CpG overlap was evaluated both before and after preprocessing to characterise the relationship between cohorts. Prior to filtering (Figure 4.11a), the intersection is dominated by probes shared between the two EPIC datasets, with a smaller but substantial core common to all three cohorts, reflecting platform design constraints and baseline annotation differences. After the complete preprocessing pipeline (Figure 4.11b), the total number of CpGs is markedly reduced, yet a large shared core is preserved: the three-way intersection comprises 215,919 CpGs, representing a stable autosomal backbone consistently retained across platforms after technical filtering.

4.5 Post-Preprocessing Feature Space and Dimensionality Challenge

The preprocessing pipeline developed in this chapter has systematically addressed the three main classes of technical confounders identified in Chapter 3: probe-design bias, unreliable probe sets, and statistical scale distortion. The resulting matrices are technically clean, fully observed, and expressed on a variance-stabilised M-value scale suitable for inferential modelling. Across all three cohorts, the three-way CpG intersection comprises 215,919 autosomal loci retained after platform-aware filtering, constituting a stable and harmonised feature backbone for cross-dataset analyses.

Yet the preprocessing outcome also sharpens the central analytical challenge.

The post-processed datasets remain extremely high-dimensional — 251,483 CpGs in GSE69914, 530,683 in GSE225845, and 486,725 in GSE287331 — and the exploratory analyses of Chapter 3 established that the Normal–Adjacent signal is focal and low-amplitude, with mean $|\Delta\beta|$ ranging from 0.009 to 0.031 across cohorts. In this regime, the overwhelming majority of retained CpGs carry no discriminative information for the contrast of interest: including them in downstream modelling would dilute the signal, inflate variance estimates, and compromise both interpretability and generalisation.

The limiting factor is therefore no longer data quality but *signal localisation*: identifying the restricted subset of CpG loci that capture biologically meaningful Normal–Adjacent variation, consistently across cohorts and independently of platform-specific noise. This requires a principled feature selection strategy that is sensitive to the subtle, focal nature of the epigenetic drift documented here, robust to the cohort-level heterogeneity observed in the inter-dataset comparisons, and compatible with the homoscedastic M-value representation adopted for modelling.

Chapter **??** addresses this challenge directly, developing and evaluating feature selection methodologies applied separately within each dataset-specific CpG space. In order to preserve the full informational richness of the EPIC platforms, feature selection is not restricted to the three-way CpG intersection but is performed on the complete post-processed feature set of each cohort, thereby avoiding artificial dimensional truncation driven by the lower coverage of the 450K array.

Chapter 5

Robust Feature Selection for Epigenetic Drift Characterisation

5.1 Statistical Challenges of High-Dimensional DNA Methylation Data

The preprocessing framework developed in Chapter 4.5 established a technically harmonised stabilised methylation space. However, preprocessing alone does not address the central statistical challenge of genome-wide DNA methylation data: extreme dimensionality relative to sample size. This challenge is particularly acute in the present study, where the objective is to characterise epigenetic differences along the Normal → Adjacent → Tumour axis in breast cancer — a setting in which biologically meaningful signal is expected to be subtle, spatially structured, and embedded in a feature space of several hundred thousand CpG loci.

High-throughput methylation arrays, such as the Illumina EPIC platform, interrogate approximately 850 000 CpG sites per sample [52], while the number of available subjects in any given cohort remains comparatively limited. This high-dimensional, low-sample-size (HDLSS) regime is well characterised in the statistical literature as a setting in which standard estimators become geometrically pathological: as dimensionality grows relative to n, pairwise distances concentrate, covariance estimation becomes unstable, and classifiers overfit to noise [53, 54]. In genomics specifically, the consequences are well documented — unconstrained feature spaces yield inflated classification accuracy that fails to replicate in independent cohorts [55, 56].

Feature selection is therefore not a cosmetic reduction step but a structural necessity for any learning procedure applied to molecular profiling data.

5.1.1 Structural Properties Motivating Feature Selection

Three properties of DNA methylation data collectively motivate explicit dimensionality control.

High Dimensionality and Overfitting Risk In HDLSS settings, discriminative models may achieve near-perfect separation on training data while capturing noise rather than biological signal. The phenomenon has been formally characterised by [53], who showed that in very high dimensions, data from distinct classes become approximately equidistant unless genuine low-dimensional structure is present. Practically, this means that feature selection must precede, not follow, model fitting — and must itself be performed exclusively on training data to avoid optimistic bias [55].

Spatial Correlation and Redundancy CpG loci exhibit strong local correlation driven by genomic proximity, chromatin organisation, and shared regulatory context. This spatial autocorrelation has been well characterised in large-scale mapping studies: contiguous CpGs within co-methylated blocks (“methylation domains”) can span hundreds to thousands of base pairs and behave as single epigenetic units [57, 58]. Without explicit redundancy control, learning algorithms may over-represent densely correlated genomic regions while underweighting distributed epigenetic signals. This is especially relevant here, as differentially methylated regions (DMRs) along the Normal–Adjacent axis are expected to be spatially clustered rather than randomly distributed [59].

Biological Sparsity Disease-associated methylation alterations are typically sparse and structured. In breast cancer, systematic analyses have demonstrated that tumour-associated hypermethylation preferentially targets specific regulatory programmes — including Polycomb-repressed developmental genes and oestrogen-regulated loci — while genome-wide methylation changes in histologically normal adjacent tissue are more subtle but nonetheless detectable [60, 61]. This sparsity motivates filtering strategies that prioritise stable, biologically coherent CpG subsets over exhaustive feature retention.

5.1.2 Methodological Challenges in Genomic Feature Selection

Feature selection in genomic data presents challenges that go beyond simple variable ranking.

Information Leakage The most consequential pitfall is information leakage: when feature selection is performed on the full dataset prior to cross-validation, the evaluation procedure has implicitly seen the test labels during variable ranking. [55] demonstrated empirically that this leakage can inflate estimated accuracy by 10–40 percentage points in gene expression data; analogous inflation has been documented for methylation-based classifiers [56]. Strict containment of all selection steps within the training fold is therefore essential and is enforced throughout the present framework.

Significance Versus Stability A second challenge concerns the relationship between statistical significance and effect stability. In large feature spaces, extremely small p-values may arise for negligible or unstable effect sizes — a manifestation of the multiple testing problem that is not fully resolved by correction alone [62]. Conversely, biologically relevant features may show moderate but reproducible signal across subsamples. Stability selection [63] addresses this by quantifying the reproducibility of feature inclusion across repeated subsampling, providing selection frequency as a complementary criterion to statistical rank.

Predictive Optimisation Versus Interpretability A third challenge is the tension between predictive optimisation and interpretability. Aggressive feature selection driven solely by discriminative performance may yield compact but biologically opaque models, while overly conservative selection may retain sufficient redundancy to obscure the underlying regulatory architecture. In translational contexts, this tension has additional implications: in-silico classification accuracy does not automatically imply clinical utility [64], and the biological coherence of the selected feature set is therefore a criterion in its own right.

5.1.3 Objectives of the Present Framework

In light of the statistical and biological considerations discussed above, the feature selection strategy developed in this chapter is designed to satisfy four interconnected objectives.

- **Strict leakage control**: all selection procedures are confined to training data, preventing inflation of evaluation metrics through inadvertent information exposure.

- **Stability enforcement**: CpGs are retained only if they demonstrate reproducible discriminative behaviour across repeated stratified resampling, ensuring robustness to training-set perturbations.

- **Redundancy reduction**: correlation-based pruning limits the over-representation of co-methylated genomic blocks and promotes independent signal components.
- **Genomic diversification**: structural constraints prevent excessive concentration of selected features within specific chromosomes or regulatory contexts.

The objective is therefore not merely dimensionality reduction, but the identification of a stable, biologically coherent, and generalisable CpG subset capable of characterising the Normal–Adjacent epigenetic transition and supporting cross-dataset validation.

5.2 General Feature Selection Framework

Building upon the statistical considerations outlined above, this section formally describes the feature selection workflow adopted in the present study. The procedure operates on the post-preprocessing methylation matrices defined in Chapter 4 and is structured as a sequence of strictly train-only operations. These steps progressively reduce dimensionality while preserving statistical validity, stability, and biological coherence. From variability screening and scale transformation to stability-based ranking and redundancy control, each component addresses a specific structural property of DNA methylation data, culminating in a diversified CpG subset suitable for downstream cross-dataset evaluation.

5.2.1 Empirical Variability-Based Dimensionality Reduction

The first stage of the feature selection framework consists of an unsupervised dimensionality reduction step aimed at removing CpGs that exhibit negligible variability within the tissue under study. This strategy is grounded in the empirically driven data reduction method proposed by [65], who demonstrated that a substantial fraction of CpGs interrogated by the Illumina 450K array are effectively non-variable within a given tissue context, contributing to the multiple testing burden — by inflating the number of hypotheses tested — without increasing the probability of detecting true differential methylation signals.

Data restriction and structural cleaning Let$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$, where $x_i \in [0,1]^p$ represents the vector of β-values for p CpG loci and $y_i \in \{0,1\}$ encodes the binary Normal–Adjacent label, with $y_i = 0$ denoting histologically normal tissue and $y_i = 1$ denoting tumour-adjacent tissue. Samples belonging to other tissue states (e.g. Tumour) are excluded from $\mathcal{D}$ prior to feature selection. This restriction ensures that the selected CpGs specifically characterise early epigenetic alterations occurring in histologically normal tissue adjacent to tumour, rather than late-stage tumour-specific methylation changes.

Train–test partition The dataset is partitioned into a training set $\mathcal{D}_{\text{train}}$ and a held-out test set $\mathcal{D}_{\text{test}}$ via stratified random sampling:

$$\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{test}}, \quad |\mathcal{D}_{\text{train}}| = 0.80\, n, \quad |\mathcal{D}_{\text{test}}| = 0.20\, n, \tag{5.1}$$

with class proportions preserved in both sets. Crucially, all feature selection statistics — including the variability filter defined below — are estimated exclusively on $\mathcal{D}_{\text{train}}$ and the resulting CpG subset is applied unchanged to $\mathcal{D}_{\text{test}}$. This design prevents information leakage and ensures that dimensionality reduction is statistically independent of evaluation [55].

Definition of the variability statistic For each CpG $j \in \{1, \ldots, p\}$, variability is quantified using the robust inter-quantile range statistic proposed by [65]:

$$r_{\beta,j} = Q_{0.90}(\beta_j \mid \mathcal{D}_{\text{train}}) - Q_{0.10}(\beta_j \mid \mathcal{D}_{\text{train}})\,, \tag{5.2}$$

where $Q_{0.90}$ and $Q_{0.10}$ denote the empirical 90th and 10th percentiles of the β-value distribution computed across all samples in $\mathcal{D}_{\text{train}}$, pooling both tissue classes. Note that $r_{\beta,j}$ measures population-level dispersion unconditional on class membership: it quantifies whether a CpG is variable across individuals at all, not whether it discriminates between tissue types. A CpG is retained if and only if:

$$r_{\beta,j} \geq \tau, \qquad \tau = 0.05, \tag{5.3}$$

yielding the retained index set $\mathcal{J}_{\text{Edgar}} = \{j \in \{1, \ldots, p\} : r_{\beta,j} \geq \tau\}$. The threshold $\tau = 0.05$ corresponds to the empirical criterion introduced in [65]: CpGs exhibiting a population-level methylation range below 5% were operationally defined as non-variable and excluded. In the original study, this filter removed approximately 20–30% of probes on the 450K array across diverse tissue types [65].

Interpretation and scope Because $r_{\beta,j}$ is computed without reference to class labels, this filtering step is strictly unsupervised: it cannot introduce label-dependent bias and is therefore safe to apply upstream of the stratified cross-validation framework described in Section 5.1.2. Its function is orthogonal to subsequent supervised ranking steps — it defines the candidate space over which discriminative selection will operate, not the final feature set.

CpGs excluded by this criterion ($r_{\beta,j} < 0.05$) typically correspond to loci consistently hypo- or hypermethylated across individuals, frequently enriched in CpG islands and constitutively silenced promoter regions [65, 66]. Retaining such loci inflates the effective dimensionality of the feature space without contributing discriminative variance, exacerbating the HDLSS geometry. Low-variance independent filtering is a standard strategy in high-dimensional genomic analyses to reduce noise and improve statistical efficiency without compromising type I error control [67].

From a statistical perspective, $|\mathcal{J}_{\text{Edgar}}|$ defines the reduced multiple testing space on which all downstream differential methylation analyses and stability selection procedures are conducted. The CpG subset $\mathcal{J}_{\text{Edgar}}$, estimated on $\mathcal{D}_{\text{train}}$, is applied identically to $\mathcal{D}_{\text{test}}$ prior to any evaluation (Equation 5.1), preserving strict train–test separation throughout.

5.2.2 Re-alignment to M-values and Residual Variance Regularisation

The empirical variability screening of Section 5.2.1 is carried out in β-space, preserving continuity with the original inter-quantile dispersion criterion of [65]. This is consistent with the dual-representation strategy of Section 4.2.4: β-values are retained for biological interpretability, while M-values are adopted for statistical modelling because their logit transformation substantially stabilises the mean–variance relationship [6].

Once $\mathcal{J}_{\text{Edgar}}$ has been determined exclusively on $\mathcal{D}_{\text{train}}$, the analysis transitions to the M-value representation defined in Equation (4.1). The transformation is applied deterministically to the post-Edgar β matrices of both training and test sets: no model parameters are estimated at this stage. For each retained CpG $j \in \mathcal{J}_{\text{Edgar}}$, using the same numerical stabilisation constant introduced during preprocessing. The CpG subset selected in β-space is transferred unchanged to M-space, preserving strict train–test separation throughout.

Residual low-variance trimming (train-only) Although the Edgar filter removes CpGs with negligible population-level dispersion in β-space, a residual fraction of near-constant loci may survive after logit transformation. This occurs because the logit map is nonlinear: loci with β-values concentrated near the boundaries of [0,1] can exhibit low inter-quantile range yet produce near-constant M-values once mapped to the real line. Retaining such loci inflates the effective dimensionality of the covariance structure without contributing discriminative signal, which is particularly detrimental in HDLSS settings.

To address this, the pipeline systematically discards the lowest α-quantile of CpGs ranked by their empirical standard deviation in M-space on $\mathcal{D}_{\text{train}}$. Letting s_j denote this standard deviation, the retained index set is

$$\mathcal{J}_{\text{var}} = \{j \in \mathcal{J}_{\text{Edgar}} : s_j \geq Q_\alpha(s)\}, \qquad \alpha = 0.02, \tag{5.4}$$

corresponding to the removal of the bottom 2% of CpGs by dispersion. The step is unsupervised — class labels play no role — and introduces no information leakage. The quantile threshold is estimated solely on $\mathcal{D}_{\text{train}}$; the resulting binary mask is then applied unchanged to $\mathcal{D}_{\text{test}}$, maintaining complete statistical independence of the evaluation set.

The resulting set $\mathcal{J}_{\text{var}}$ constitutes the final candidate feature space entering the supervised discriminative ranking stage described in Section 5.1.2.

5.2.3 Biologically Weighted Stability Selection and Region-Level Consolidation

After structural dimensionality control (Sections 5.2.1 and 5.2.2), the feature space remains in the order of several hundred thousand CpGs. In HDLSS regimes, univariate ranking alone is insufficient: small perturbations in training composition can induce large instability in feature ordering, particularly when signal-to-noise ratios are moderate [53, 54]. To address this, we adopt a resampling framework that integrates statistical evidence, directional consistency, and biologically informed weighting across four sequential components.

Repeated Stratified Stability Framework Let $X \in \mathbb{R}^{n_{\text{train}} \times p}$ denote the post-filtered M-value matrix and $y \in \{0,1\}^{n_{\text{train}}}$ the binary Normal–Adjacent label. A Repeated Stratified K-Fold (RSKF) procedure is employed with $K = 5$ folds and $R = 10$ repetitions, yielding $S = 50$ train/validation splits. Repeated stratified resampling reduces the variance of estimated feature importance and mitigates selection instability arising from dependence on a single partition [55, 56].

For each split $s = 1, \ldots, S$ and each CpG j, three quantities are computed exclusively on the training fold:

1. The signed mean difference $\Delta M_j^{(s)} = \bar{M}_{1,j}^{(s)} - \bar{M}_{0,j}^{(s)}$ is computed for all CpGs as an effect-size measure.

2. To reduce computational burden in the HDLSS regime, Mann–Whitney U p-values are evaluated only for the top 50,000 CpGs ranked by $|\Delta M_j^{(s)}|$ within the training fold. CpGs outside this prefiltered subset are assigned $p_j^{(s)} = 1$, corresponding to the least favourable rank. This two-stage procedure introduces a structural dependency between the prescreening criterion and the subsequent rank aggregation, since excluded CpGs receive a conservative rank by construction rather than by statistical evidence. The approximation is justified by the width of the prefilter: retaining the top 50,000 candidates corresponds to approximately 6% of the full feature space and substantially exceeds the final selection target of 5,000 CpGs, thereby limiting the probability

of excluding genuinely discriminative loci from downstream evaluation. The Mann–Whitney test is adopted as a rank-based nonparametric procedure to avoid distributional assumptions and ensure robustness to residual deviations from normality in M-space [68].

3. The same signed difference is computed independently on the validation fold to assess directional reproducibility across the partition.

Two rank statistics are then derived per split, $r_{p,j}^{(s)}$ (ascending rank of $p_j^{(s)}$), $r_{|\Delta M|,j}^{(s)}$ (descending rank of $|\Delta M_j^{(s)}|$), and averaged across splits:

$$\overline{r}_{p,j} = \frac{1}{S}\sum_{s=1}^{S} r_{p,j}^{(s)}, \qquad \overline{r}_{\Delta M,j} = \frac{1}{S}\sum_{s=1}^{S} r_{|\Delta M|,j}^{(s)}. \tag{5.5}$$

The combined statistical score is the equal-weight convex combination:

$$\text{score}_{\text{stat},j} = 0.5\,\overline{r}_{p,j} + 0.5\,\overline{r}_{\Delta M,j}. \tag{5.6}$$

This rank aggregation follows a Borda-count-like scheme [69], in which equal-weight combination of complementary ranking criteria has been shown to reduce selection variance relative to either criterion alone [70] The resulting statistical score is subsequently min–max normalised to the unit interval:

$$\text{score}_{\text{stat},j}^{\text{norm}} = \frac{\text{score}_{\text{stat},j} - \min(\text{score}_{\text{stat}})}{\max(\text{score}_{\text{stat}}) - \min(\text{score}_{\text{stat}})}.$$

Relying solely on p-values risks prioritising negligible but stable effects under large sample sizes, whereas magnitude-only ranking may overweight noisy large deviations. Combining both criteria mitigates this trade-off and is consistent with resampling-based stability principles in high-dimensional feature selection [63].

Directional Stability Constraint A CpG whose estimated effect direction reverses across resampling splits is unlikely to reflect a genuine, reproducible biological signal. We therefore impose a directional reproducibility criterion. For each CpG j, let $\text{sign}_{\text{train}}^{(s)}$ and $\text{sign}_{\text{val}}^{(s)}$ denote the signs of $\Delta M_j^{(s)}$ on the training and validation folds of split s, respectively. The directional stability score is:

$$\text{stab}_j = \frac{\#\left\{s : \text{sign}_{\text{train}}^{(s)} = \text{sign}_{\text{val}}^{(s)} \neq 0\right\}}{\#\left\{s : \text{sign}_{\text{train}}^{(s)} \neq 0 \text{ and } \text{sign}_{\text{val}}^{(s)} \neq 0\right\}}. \tag{5.7}$$

The directional stability score is computed as the proportion of resampling splits in which both training and validation folds exhibit non-zero effect direction and agree in sign. CpGs for which the denominator equals zero — i.e. all splits yield $\Delta M_j^{(s)} = 0$ on at least one fold — are assigned $\text{stab}_j = 0$ by convention, rendering them ineligible for selection under the stability constraint. Only CpGs with $\text{stab}_j \geq 0.75$ are retained, requiring directional agreement in at least 75% of the splits for which a non-zero effect direction is observed in both folds. The threshold of 0.75 is chosen in analogy with the selection frequency cutoff recommended by [63], where a 75% inclusion rate across subsampling splits is identified as a conservative criterion for stable feature identification. The same numerical threshold is here applied to directional consistency rather than selection frequency, extending the stability principle from feature inclusion to effect reproducibility — a particularly relevant criterion when the target signal is moderate in magnitude, as expected for early epigenetic drift in histologically normal adjacent tissue.

Adaptive Effect-Size Filtering A global effect estimate is computed on the full training set, $\Delta M_j^{\text{full}} = \bar{M}_{1,j} - \bar{M}_{0,j}$, and a CpG is retained only if $|\Delta M_j^{\text{full}}| \geq Q_{0.60}(|\Delta M^{\text{full}}|)$. Rather than applying a fixed ΔM cutoff — which would be sensitive to inter-cohort differences in dispersion — this adaptive percentile-based criterion maintains scale invariance across datasets. A fallback to the 50th percentile is applied when the resulting candidate pool would otherwise be insufficient for downstream selection. Together with the directional stability constraint, this filter ensures that only CpGs with both reproducible direction and non-negligible magnitude reach the biological weighting stage. The fallback to the 50th percentile is triggered whenever the resulting candidate pool contains fewer than 15,000 CpGs, ensuring sufficient capacity for the downstream consolidation stage.

Integration of Biological Priors Purely statistical ranking may underweight loci in regulatory regions known to mediate transcriptional control. Promoter-associated CpGs and CpG islands are well-established sites of regulatory methylation changes in cancer and early field effects [57, 61]. Each CpG is assigned a discrete biological prior weight $w_j \in \{1.00, 0.70, 0.50, 0.20\}$ according to its island and promoter annotation:

$$w_j = \begin{cases} 1.00 & \text{Island + TSS} \\ 0.70 & \text{Island only} \\ 0.50 & \text{Promoter (TSS) only} \\ 0.20 & \text{Other loci} \end{cases} \tag{5.8}$$

The discrete weights are subsequently min–max normalised to the unit interval, yielding $w_j^{\text{norm}} \in [0,1]$. In order to prioritise biologically annotated loci while preserving a lower-is-better scoring convention, the biological contribution is defined as $1 - w_j^{\text{norm}}$.

After min–max normalisation of the statistical score $\text{score}_{\text{stat},j}$, the final integrated score is

$$\text{score}_j = (1 - \lambda)\,\text{score}^{\text{norm}}_{\text{stat},j} + \lambda\,(1 - w^{\text{norm}}_j), \qquad \lambda = 0.20. \tag{5.9}$$

Min–max normalisation is applied independently to each component prior to combination, ensuring commensurability of scale. The procedure is sensitive to extreme values; however, given the large number of CpGs ($|\mathcal{J}_{\text{var}}| \gg 10^4$), the influence of individual outliers on the normalised distribution is expected to be negligible. The coefficient $\lambda = 0.20$ reflects a deliberate design choice rather than an empirically optimised parameter: statistical evidence contributes 80% of the integrated score, while the remaining 20% provides modest upweighting for loci with established regulatory function in promoter methylation and transcriptional silencing [71]. The top 15,000 CpGs satisfying both the stability and adaptive effect-size constraints define the candidate feature pool passed to the next stage.

Region-Level Anchoring on CpG Islands CpGs are not independent genomic entities: spatially proximal loci within co-methylated blocks co-vary as coordinated regulatory units [57]. Selecting isolated top-ranked probes therefore risks capturing stochastic single-probe fluctuations rather than coherent epigenetic events, which is particularly problematic when the signal of interest — early field-effect drift — is expected to manifest as concerted changes across regulatory domains.

To enforce spatial coherence, candidate CpGs are grouped by annotated CpG island. For each island containing at least two candidate CpGs, a region-level score is defined as:

$$\text{region\_score} = \text{median}(\text{score}_j) \times \text{fraction}_{\text{stable}}, \tag{5.10}$$

where $\text{fraction}_{\text{stable}}$ denotes the proportion of CpGs within the island satisfying both $\text{stab}_j \geq 0.75$ and $|\Delta M^{\text{full}}_j| \geq \tau$, with τ denoting the adaptive effect-size threshold selected in the previous step (60th percentile or 50th percentile fallback). The multiplicative formulation jointly penalises islands in which either the median discriminative score is weak or the proportion of internally stable CpGs is low: an island driven by isolated high-scoring probes surrounded by unstable loci receives a substantially discounted region score, ensuring that anchored islands exhibit both statistical quality and internal coherence. Only islands with $\text{fraction}_{\text{stable}} \geq 0.60$ and at least two candidate CpGs are retained. From each such island, the top $m = 2$–3 CpGs are selected into an anchored pool; the remaining candidates form a complementary non-anchored set. This hierarchical consolidation reduces susceptibility to single-probe artefacts and aligns feature selection with the biological expectation that early epigenetic drift occurs in clustered regulatory domains rather than as isolated stochastic events.

5.2.4 Correlation-Based Redundancy Pruning and Graph-Theoretic Clustering

Despite stability filtering and region-level anchoring, the candidate feature space retains strong local correlation. DNA methylation levels exhibit pronounced spatial autocorrelation driven by genomic proximity, shared chromatin context, and coordinated regulatory control [57, 58]: in high-density arrays such as EPIC, neighbouring CpGs within islands and promoter regions frequently co-vary as near-redundant blocks. Retaining multiple highly correlated loci inflates the effective dimensionality of the design matrix without increasing independent information content, inducing multicollinearity, unstable coefficient estimates, and variance inflation in downstream classifiers [54, 53]. Explicit correlation-based redundancy pruning is therefore applied to the candidate pool before passing it to the diversification stage.

Correlation Structure on Training Data Let $Z \in \mathbb{R}^{n_{\text{train}} \times q}$ denote the M-value matrix restricted to the q candidate CpGs, with each column standardised using training-set mean μ_j and standard deviation s_j:

$$Z_{ij} = \frac{M_{ij} - \mu_j}{s_j}.$$

where s_j is the sample standard deviation computed with Bessel correction (ddof = 1) on the training set. A small numerical constant is added in practice to prevent division by zero in degenerate cases. The empirical Pearson correlation matrix is then:

$$C = \frac{1}{n_{\text{train}} - 1} Z^\top Z. \tag{5.11}$$

Because columns of Z are standardised to zero mean and unit sample variance, Equation (5.11) coincides with the empirical Pearson correlation matrix estimated exclusively on $\mathcal{D}_{\text{train}}$. All quantities are estimated exclusively on $\mathcal{D}_{\text{train}}$ to preserve strict evaluation independence.

Graph Construction and Connected Components An undirected graph $G = (V, E)$ is constructed with one vertex per candidate CpG; an edge connects j and k whenever their absolute correlation exceeds the threshold:

$$(j, k) \in E \iff |C_{jk}| \geq \tau_c, \qquad \tau_c = 0.85, \tag{5.12}$$

consistent with strong-correlation redundancy thresholds commonly adopted in high-dimensional omics analyses [72, 73]. By convention, each vertex is connected to itself in the adjacency matrix; this does not alter the connected component structure but simplifies implementation. Clusters are defined as the connected components of G: a cluster $\mathcal{C}$ is a maximal vertex subset in which every pair (j, k) is joined by at least one path in G.

This graph-theoretic formulation is strictly preferable to pairwise sequential pruning because correlation propagates transitively: two CpGs with $|C_{jk}| < \tau_c$ may both be strongly correlated to a third, and would therefore represent the same co-methylated block without being linked directly. Connected components capture this transitive redundancy structure in a single, consistent pass.

Representative Selection per Cluster For each connected component $\mathcal{C}_m$, a single representative CpG is retained:

$$j^* = \arg\min_{j \in \mathcal{C}_m} \text{score}_j, \tag{5.13}$$

where score_j is the integrated biologically weighted stability score of Section 5.2.3, in which a lower value denotes stronger statistical and biological evidence. CpGs lacking a valid stability score are assigned an infinite value and are therefore ineligible for selection as cluster representatives. Selecting the minimum thus retains the most discriminative and stable CpG within each correlated block, discarding all redundant co-varying loci. While region-level anchoring leverages genomic annotation to enforce spatial coherence based on CpG island structure, the present clustering step operates purely on empirical covariance structure estimated from the training data.

Separate Clustering of Anchored and Non-Anchored Pools To preserve the island-level spatial structure established by region-level anchoring before applying global redundancy reduction, clustering is performed separately on the region-anchored pool A (CpGs selected via island consolidation) and the non-anchored complementary pool B. Let $\mathcal{R}_A$ and $\mathcal{R}_B$ denote the representative sets obtained from each pool. Representatives from A are prioritised in the final union, and the redundancy-pruned feature set is: $\mathcal{R} = \mathcal{R}_A \cup \mathcal{R}_B$.

Statistical and Biological Rationale From a bias–variance standpoint, removing highly correlated predictors reduces estimator variance while preserving most of the predictive signal, since strongly correlated CpGs encode largely overlapping epigenetic information. In HDLSS settings this is particularly consequential: redundant features exacerbate the ill-conditioning of the design matrix, inflate effective model complexity, and destabilise decision boundaries [54]. Removing redundant predictors reduces effective dimensionality and improves the numerical conditioning of the design matrix without materially reducing predictive information content.

The result is a structurally compact, non-redundant feature set that retains statistical strength, directional stability, biological coherence, and spatial structure — forming the input to the genomic diversification stage described in Section 5.2.5.

5.2.5 Genomic Diversification via Constrained Greedy Selection

After correlation-based redundancy pruning (Section 5.2.4), the feature set $\mathcal{R}$ remains enriched for highly ranked CpGs but may still exhibit structural imbalance. Genome-wide methylation arrays are inherently non-uniform — CpG density varies substantially across chromosomes, islands, shores, shelves, and open sea regions — so unconstrained greedy ranking by statistical score tends to over-concentrate features in CpG-dense promoter islands, chromosomes with strong signal, and local genomic windows of coordinated drift. Such concentration artificially reduces effective genomic coverage and increases the risk that downstream models capture locus-specific rather than system-level epigenetic patterns, reducing structural generalisability across cohorts. We therefore introduce a diversification step formulated as a constrained subset selection problem.

Problem Formulation Let $\mathcal{R} = \{j_1, \ldots, j_m\}$ denote the redundancy-pruned CpG pool ordered by increasing integrated score score_j (Section 5.2.3). CpGs lacking valid manifest annotation for chromosome and genomic position are excluded prior to diversification, as spatial constraints require coordinate information. The goal is to select a subset $\mathcal{S} \subset \mathcal{R}$, $|\mathcal{S}| = K = 5000$, that maximises overall discriminative quality under structural diversity constraints. In practice, the effective target is defined as $K_{\text{eff}} = \min(K, |\mathcal{R}|)$ to accommodate cases in which fewer than K mapped CpGs are available. All constraint budgets are computed with respect to K_{eff}.

Formally:

$$
\begin{aligned}
\min_{\mathcal{S} \subseteq \mathcal{R}} \quad & \sum_{j \in \mathcal{S}} \text{score}_j \\
\text{s.t.} \quad & |\mathcal{S}| = K_{\text{eff}}, \\
& |\{j \in \mathcal{S} : c(j) = \chi\}| \leq \lfloor \alpha_{\text{chr}} K_{\text{eff}} \rfloor \quad \forall \chi, \\
& |\{j \in \mathcal{S} : \gamma(j) = g\}| \leq \lfloor \alpha_{\text{ctx}} K_{\text{eff}} \rfloor \quad \forall g, \\
& |\{j \in \mathcal{S} : w(j) = \omega\}| \leq \beta \quad \forall \omega.
\end{aligned}
\tag{5.14}
$$

Each constraint targets a distinct axis of potential structural imbalance and is described in turn below.

Chromosomal Balance Constraint Let $c(j)$ denote the chromosome of CpG j. No chromosome may contribute more than a fixed fraction of the final set:

$$|\{j \in \mathcal{S} : c(j) = \chi\}| \leq \alpha_{\mathrm{chr}} K, \qquad \alpha_{\mathrm{chr}} = 0.08. \tag{5.15}$$

The budget per chromosome is computed as $\lfloor \alpha_{\mathrm{chr}} K_{\mathrm{eff}} \rfloor$, with a minimum of one CpG allowed per chromosome. The 8% cap prevents over-representation of chromosomes that harbour dense signal regions, a configuration known to affect classifier stability when disease-associated loci cluster in specific chromosomal domains [72].

Genomic Context Constraint Let $\gamma(j)$ denote the CpG island context of locus j (Island, Shore, Shelf, or OpenSea). The fraction of any single context class is bounded by:

$$|\{j \in \mathcal{S} : \gamma(j) = g\}| \leq \alpha_{\mathrm{ctx}} K, \qquad \alpha_{\mathrm{ctx}} = 0.65. \tag{5.16}$$

The context budget is computed as $\lfloor \alpha_{\mathrm{ctx}} K_{\mathrm{eff}} \rfloor$, with at least one CpG permitted per context class. Although promoter islands are biologically important, cancer-related methylation changes are not confined to islands and frequently involve shores and open sea regions [57, 61]. The soft cap ensures that non-island regulatory regions retain representation in the final panel.

Spatial Window Constraint To prevent local hyper-concentration, each chromosome is partitioned into non-overlapping windows of $W = 500\,\mathrm{kb}$, and the number of selected CpGs per window is hard-capped: $|\{j \in \mathcal{S} : w(j) = \omega\}| \leq 15$, where $w(j)$ is the window index of CpG j. Formally, windows are defined via integer binning $w(j) = \lfloor \mathrm{pos}(j)/W \rfloor$ within each chromosome. Since methylation domains frequently span tens to hundreds of kilobases [57], this constraint preserves regional representation while preventing dominance by a single hyper-variable block — a risk that correlation pruning alone cannot fully eliminate at the macro-scale.

Greedy Approximation with Progressive Relaxation The constrained optimisation problem (5.14) is combinatorial and NP-hard in general, as it resembles cardinality-constrained subset selection with multiple linear constraints. Rather than solving a full mixed-integer program, we adopt a deterministic greedy strategy: CpGs are processed in order of increasing score_j, ties are resolved deterministically by stable ordering, and each candidate is added to $\mathcal{S}$ if and only if all active constraints are simultaneously satisfied, until $|\mathcal{S}| = K$.

If the strict constraint regime fails to reach K, constraints are relaxed by staged deactivation (rather than by increasing thresholds), first removing the spatial window constraint, then the genomic context constraint, and finally the chromosome balance constraint.

At each stage, the specified constraint is fully deactivated while previously relaxed constraints remain inactive. Spatial over-concentration is relaxed first because local genomic crowding is most detrimental to coverage; chromosomal imbalance is relaxed last as it has the weakest direct effect on independent information content. The constrained greedy strategy does not admit formal approximation guarantees in this setting, as the objective is linear rather than submodular and the constraint structure involves multiple simultaneous bounds. The approach is adopted as a computationally tractable heuristic; its empirical adequacy is assessed indirectly through the constraint activation frequency reported in Section 5.3. The frequency with which progressive constraint relaxation is activated in practice is reported in Section 5.3 for each dataset, allowing empirical assessment of whether the final selection approximates the fully constrained solution or degrades toward unconstrained score-based ranking.

5.3 Dataset-Specific Implementation

5.3.1 Dataset GSE69914

5.3.2 Dataset GSE225845

5.3.3 Dataset GSE287331

5.4 Inter-Dataset Consistency After Feature Selection

5.4.1 Cross-Cohort Overlap of Selected CpGs

5.4.2 Transferability of Stability Scores Across Cohorts

5.4.3 Concordance of Directional Epigenetic Drift

5.5 Synthesis and Transition to Predictive Modelling

5.5.1 Structural Properties of the Final CpG Panel

5.5.2 Implications for Supervised Learning and Model Design

Chapter 6

Model training and evaluation

REPORT 04

Qui ci va - Classificatori - Problema dello zaino - Gene set analysis - Problemi finali per giustificare cap7 di CpGPT (spero)

Chapter 7

CpGPT - estensione cross dataset

Chapter 8

Conclusion and future work

CONCLUSIONI E PROSPETTIVE DI FUTURO

Appendix A

Filtering lists

A.1 Probe Filtering Resources

This appendix documents the five external annotation resources adopted for technical probe filtering across all three datasets. Each resource targets distinct classes of measurement artefacts; together they define the union-based exclusion criterion applied in Section 4.2.3. Table A.1 summarises the technical categories covered by each resource.

A.1.1 Naeem *et al.* (2014)

Naeem *et al.* [40] proposed a structured filtering framework for the HumanMethylation450 array, designed to reduce false discoveries by sequentially discarding probes on the basis of hybridisation specificity and genomic integrity. The framework excludes probes that multi-map or cross-hybridise to additional genomic loci, probes overlapping repetitive elements (LINE, SINE, Alu), and probes overlapping regions harbouring insertion–deletion polymorphisms (INDELs). For SNP-affected probes, a distinction is drawn between variants at the interrogated CpG or single-base extension site—which directly compromise the methylation measurement and are therefore excluded—and variants located nearby but not interfering in bisulfite space, which are retained. The resulting exclusion set thus removes probes affected by cross-reactivity, structural polymorphisms, and measurement-critical SNPs.

A.1.2 Chen *et al.* (2013)

Chen *et al.* [41] empirically identified probes in the 450K array exhibiting off-target hybridisation or overlap with common SNPs. In the present work, only the cross-reactive probe list is used. This catalogue enumerates approximately 29,000 multi-mapping probes whose methylation signal cannot be unambiguously

Table A.1: Technical artefact categories covered by each filtering resource.

Category	Naeem	Chen	Pidsley	Zhou	McCartney
Cross-hybridisation	Yes	Yes	Yes	`MASK_mapping`	Yes
SNP at CpG	Yes	—	Yes	`MASK_snp5`, `MASK_extBase`	—
Nearby SNP	Conditional	—	—	—	—
Structural variant	Yes	—	—	—	—
Non-CpG probes	—	—	Yes	`MASK_nonCG`	Yes

attributed to a single genomic locus, and whose inclusion would therefore introduce spatially unresolved signal into the feature matrix.

A.1.3 Pidsley *et al.* (2016)

Pidsley *et al.* [19] provide a platform-level assessment of the EPIC array, releasing annotated probe lists that flag loci where technical bias arises from design- or genome-related sources. Three categories are relevant here. First, CpG-targeting probes with sequence homology to additional genomic loci, which yield non-unique hybridisation and potentially ambiguous β estimates. Second, non-CpG-targeting probes affected by analogous off-target issues, excluded due to limited interpretability in CpG-centric analyses. Third, probes overlapping common genetic variation at three critical positions: the interrogated CpG itself (polymorphic CpG, directly affecting the presence of the CpG dinucleotide); the single-base extension site of Type I probes (perturbing extension chemistry and dye-intensity ratios); and within the probe body (reducing binding affinity or altering hybridisation kinetics for common variants).

A.1.4 Zhou *et al.* (2016)

Zhou *et al.* [18] released a unified probe annotation for both 450K and EPIC arrays encoding technical reliability through a set of binary mask columns (`MASK_*`). The masks applied in this work flag probes with low or inconsistent mapping quality (`MASK_mapping`); Type I probes carrying a SNP at the extension base that induces a colour-channel switch (`MASK_typeINextBaseSwitch`); probes whose extension base is inconsistent with the expected channel or CpG context (`MASK_extBase`); probes with non-unique 30-bp 3′ subsequences susceptible to cross-hybridisation (`MASK_sub30.copy`); probes overlapping an SNP within ± 5 bp of the interrogated CpG (`MASK_snp5.common`); and probes overlapping SNPs with global minor allele frequency above 1% (`MASK_snp5.GMAF1p`). The composite `MASK_general` field, which integrates mapping, SNP, and cross-reactivity criteria, is used as the primary filter for general-purpose exclusion. Probes overlapping RepeatMasker regions (`MASK_rmsk15`) are not excluded, consistent with standard practice.

A.1.5 McCartney *et al.* (2016)

McCartney *et al.* [42] assessed probe design artefacts on the EPIC array and published supplementary tables enumerating cross-hybridising CpG-targeting probes and cross-hybridising non-CpG-targeting probes. These lists complement the Pidsley and Zhou resources by providing an independent empirical characterisation of off-target hybridisation on the EPIC platform.

The combined application of these resources ensures that downstream analyses operate on a probe set with maximised hybridisation specificity and minimal confounding by common genetic variation.

Appendix B

Extension of an Empirically Driven Variability-Based Filtering Framework to Breast Tissue

B.1 Motivation and Scope

The variability-based filtering strategy embedded in the preprocessing pipeline of this thesis builds directly on the method proposed by [65], who demonstrated that a substantial fraction of CpG sites interrogated by the Illumina HumanMethylation450K array are effectively non-variable within a given tissue context and can therefore be removed prior to downstream analysis without loss of discriminative information. The original study was conducted on three healthy reference tissues — blood, buccal epithelial cells, and placenta — and the resulting non-variable CpG lists were shown to be robust across preprocessing pipelines and normalisation strategies.

The present appendix documents the extension of this framework to breast tissue. The extension pursues two objectives: (i) to assess whether the non-variability criterion, as originally defined, generalises to a cancer-relevant tissue under increased biological heterogeneity; and (ii) to provide a general, reproducible protocol for incorporating additional tissues into the reference framework in future studies. The analysis operates exclusively on histologically normal samples to preserve comparability with the original reference tissues and to estimate baseline methylation stability in the absence of disease-related confounding. Importantly, this appendix does not modify the non-variability definition of [65]: a CpG is classified as non-variable if its inter-quantile range $r_{\beta,j} = Q_{0.90}(\beta_j) - Q_{0.10}(\beta_j) < 0.05$. The contribution is therefore evaluative and integrative rather than definitional.

B.2 General Framework for Cross-Tissue Extension

Let $\mathcal{T} = \{T_1, \ldots, T_k\}$ denote a set of reference tissues, each represented by one or more independent methylation datasets from normal samples. For each tissue T_ℓ, the tissue-specific non-variable CpG set is defined as:

$$\mathcal{N}_\ell(\tau) = \left\{ j \in \{1, \ldots, p\} : Q_{0.90}\left(\beta_j^{(\ell)}\right) - Q_{0.10}\left(\beta_j^{(\ell)}\right) < \tau \right\}, \tag{B.1}$$

where $\tau = 0.05$ is the threshold inherited from [65]. The cross-tissue invariant core — CpGs non-variable across all reference tissues simultaneously — is given by the intersection:

$$\mathcal{N}_{\text{core}}(\tau) = \bigcap_{\ell=1}^{k} \mathcal{N}_\ell(\tau). \tag{B.2}$$

When a new tissue T_{k+1} is incorporated, the invariant core is updated as:

$$\mathcal{N}'_{\text{core}}(\tau) = \mathcal{N}_{\text{core}}(\tau) \ \cap \ \mathcal{N}_{k+1}(\tau), \tag{B.3}$$

which is strictly more conservative than the original core: $\mathcal{N}'_{\text{core}}(\tau) \subseteq \mathcal{N}_{\text{core}}(\tau)$.

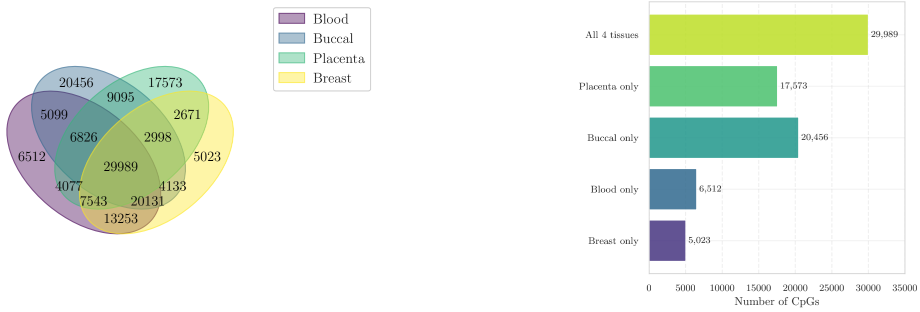
The validation of any extension requires verifying three properties before the updated core can be considered methodologically sound:

1. **Selection stability**: the non-variable set $\mathcal{N}_{k+1}(\tau)$ must be robust to dataset composition, i.e. not driven by any single cohort (Step A, Section B.3.1).
2. **Threshold optimality**: the chosen τ must lie at a favourable stability–dimensionality trade-off point in the new tissue context (Step B, Section B.3.2).
3. **Empirical coherence**: the selected CpGs must exhibit methylation and variability profiles consistent with the non-variability interpretation, ruling out selection artefacts (Steps C–D, Sections B.3.3–B.3.4).

This three-step validation protocol is general and applies identically to any future tissue extension.

B.3 Cross-Tissue Overlap Structure

Breast tissue was introduced alongside the three original reference tissues of [65]. Figure B.1 reports the overlap structure of tissue-specific non-variable CpG sets across all four tissues.



(a) Pairwise and higher-order overlaps of tissue-specific non-variable CpG sets.

(b) Quantitative intersection sizes for key subsets.

Figure B.1: Global overlap structure of non-variable CpGs across blood, buccal epithelial cells, placenta, and breast tissue.

The four-way invariant core $\mathcal{N}'_{\text{core}}(0.05)$ comprises 29,989 CpGs — a substantial shared subset whose persistence despite the inclusion of a cancer-relevant tissue suggests that at least a component of CpG non-variability reflects stable, context-independent properties of the methylation landscape rather than purely tissue-specific regulatory programmes. The breast-exclusive non-variable set (5,023 CpGs) is the smallest tissue-specific component, consistent with the expectation that breast tissue introduces additional heterogeneity relative to the original healthy tissue panel.

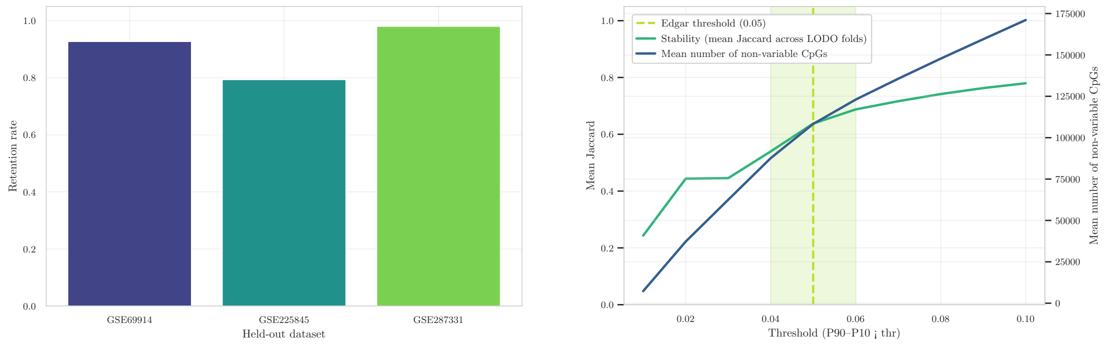
B.3.1 Step A: Stability of Non-Variable CpG Selection

The first validation step assesses whether the breast-tissue non-variable set $\mathcal{N}_{\text{breast}}(\tau)$ is robust to dataset composition. A leave-one-dataset-out (LODO) strategy is adopted: for each held-out dataset $\mathcal{D}^{(-s)}$, the non-variable set is recomputed on the remaining datasets and compared to the full-reference set via the retention rate:

$$\rho^{(-s)} = \frac{\left|\mathcal{N}^{(-s)}_{\text{breast}}(\tau) \cap \mathcal{N}_{\text{breast}}(\tau)\right|}{|\mathcal{N}_{\text{breast}}(\tau)|}, \tag{B.4}$$

where $\mathcal{N}^{(-s)}_{\text{breast}}(\tau)$ denotes the non-variable set estimated without dataset s.

The retention rates reported in Figure B.2a are consistently high across all three held-out datasets, demonstrating that $\mathcal{N}_{\text{breast}}(\tau)$ is not an artefact of any individual cohort but reflects a stable property of the underlying methylation variability structure in breast tissue. This result satisfies the first validation criterion stated in Section B.2 and is a necessary condition for interpreting non-variability as a genuine empirical property rather than a dataset-specific anomaly.



(a) Retention rates under the LODO protocol across held-out datasets (GSE69914, GSE225845, GSE287331).

(b) Mean Jaccard similarity $J(\tau)$ and mean non-variable CpG count as functions of the threshold.

Figure B.2: Assessment of robustness and trade-off between stability and dimensionality of non-variable CpG sets.

B.3.2 Step B: Stability–Dimensionality Trade-Off

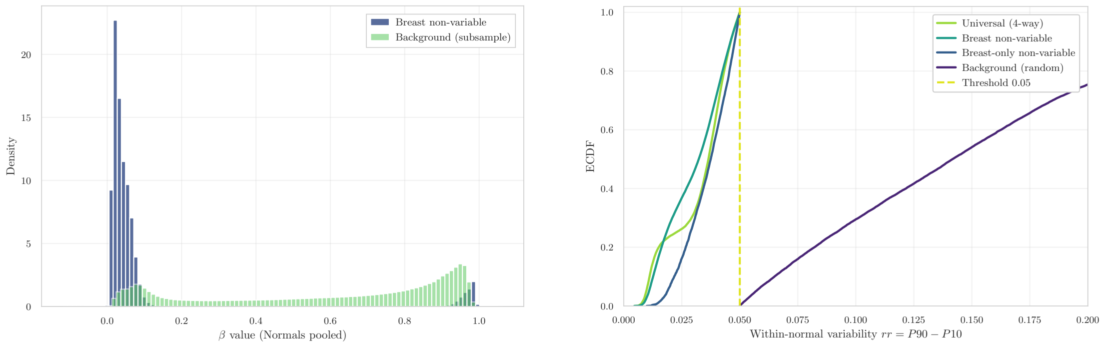
Having established stability at $\tau = 0.05$, Step B evaluates whether this threshold represents an optimal compromise between selection reproducibility and dimensionality reduction in the breast tissue setting. The mean Jaccard similarity across LODO folds and the mean number of non-variable CpGs are computed as functions of $\tau \in [0.01, 0.10]$:

$$J(\tau) = \frac{1}{S}\sum_{s=1}^{S} \frac{\left|\mathcal{N}^{(-s)}_{\text{breast}}(\tau) \cap \mathcal{N}_{\text{breast}}(\tau)\right|}{\left|\mathcal{N}^{(-s)}_{\text{breast}}(\tau) \cup \mathcal{N}_{\text{breast}}(\tau)\right|}. \tag{B.5}$$

As shown in Figure B.2b, $J(\tau)$ increases rapidly for small τ and saturates before the Edgar threshold, while the CpG count continues to grow approximately linearly. At $\tau = 0.05$, stability is already at its near-maximum value and the marginal gain from increasing the threshold further is negligible relative to the dimensionality cost incurred. This empirically validates the original threshold of [65] in the breast tissue context and provides a data-driven justification for its retention within the extended framework.

B.3.3 Step C: Methylation Distribution of Non-Variable CpGs

Step C characterises the methylation profiles of the selected non-variable CpGs by comparing their β-value distribution to a random background sample of array CpGs, computed on normal breast samples pooled across datasets.



(a) Beta-value density for non-variable CpGs (breast tissue) versus genomic background.

(b) ECDF of within-normal variability $r_{\beta,j} = Q_{0.90} - Q_{0.10}$ across CpG subsets.

Figure B.3: Characterization of the selected non-variable CpG set in terms of methylation distribution and baseline risk behavior.

Non-variable CpGs in breast tissue are strongly concentrated at extreme β values, consistent with constitutive hypo- or hypermethylated states. This pattern replicates the findings of [65] in healthy tissues and extends them to a cancer-relevant context, supporting the interpretation that non-variability reflects stable regulatory configurations rather than artefacts of tissue composition or disease-related heterogeneity. The enrichment of non-variable CpGs in CpG islands and constitutively silenced promoter regions, documented in the original study, is consistent with this bimodal profile [65, 66].

B.3.4 Step D: Baseline Variability Profile

Step D provides quantitative validation of the filtering criterion by examining the empirical cumulative distribution function (ECDF) of $r_{\beta,j}$ across four CpG subsets: the four-way shared non-variable core $\mathcal{N}'_{\text{core}}(0.05)$, the breast-specific non-variable set $\mathcal{N}_{\text{breast}}(0.05)$, the breast-exclusive component, and a random background set.

The ECDFs in Figure B.3b show a clear separation: the non-variable subsets are sharply concentrated at low $r_{\beta,j}$ values, with the majority of CpGs well below $\tau = 0.05$, whereas the background distribution spans a substantially wider range. This confirms that the selected CpGs are empirically distinct in their variability profile and not merely defined as non-variable by construction. The consistency of this pattern across all non-variable subsets — including the breast-exclusive component — further supports the robustness of the extended framework.

B.4 Positioning Within the Thesis and Generalisability

The extended framework described in this appendix provides the empirical grounding for the variability filter applied in Chapter **??**. The four validation steps collectively establish that the non-variability criterion of [65] is stable, threshold-optimal, and biologically coherent when applied to breast tissue — necessary conditions for its use as a preprocessing step in the present study.

In the current analysis, datasets are processed independently and the non-variability threshold is estimated within each cohort separately, rather than enforcing a single shared CpG list across datasets. This design preserves dataset-specific methylation structure and avoids premature cross-cohort harmonisation, while the validation performed here ensures that the filtering criterion rests on population-level properties rather than cohort-specific artefacts.

The general framework formalised in Section B.2 — update rule (Equation B.3), LODO stability assessment (Equation B.4), and Jaccard-based threshold analysis (Equation B.5) — is structurally independent of tissue type and platform density. It therefore remains applicable to future tissue extensions and to higher-density arrays such as the Illumina EPIC platform, whose expanded CpG coverage approximately doubles that of the 450K array [52].

Appendix C

Principi teorici matematici

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